

In Hot Water: Clarifying the Price Effects of Brexit*

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Abstract

Non-trade-policy barriers are central to explaining trade patterns and motivate deep trade agreements, yet their magnitude and effects remain difficult to measure. We quantify how non-trade-policy barriers between the EU and the UK changed following the introduction of the Trade and Cooperation Agreement (TCA). Focusing on the bottled water industry, where formal trade policy did not change, we find that UK consumer prices of imported bottled water rose by 17% relative to prices of the same products in continental Europe. We link this price increase to the TCA by showing that freight rates for road transport on UK-bound routes also rose by 30% relative to non-UK routes. To quantify the contribution of higher transport costs to higher prices and costs, we turn to a partial equilibrium model of the bottled water industry. We estimate that the increase in freight rates following the TCA accounts for 30% of the rise in UK consumer prices and costs for imported bottled water. These findings illustrate how trade relations may fragment markets even when explicit trade policy tools are unaffected.

JEL codes: F13, F15, F61, L66, R41

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1 Introduction

Recent years have been marked by isolationist sentiments (Colantone, Ottaviano, and Stanig, 2022). In response, countries have engaged in tariff wars or, in the case of the United Kingdom (UK), have turned away from deep trade agreements. Whereas the effects of tariff hikes are well understood, much less is known about the consequences of reversing deep trade agreements and increasing non-trade-policy barriers (Head and Mayer, 2014; Goldberg and Pavcnik, 2016).

This is for three reasons. First, whereas data on trade-policy tools such as tariffs and quotas are ubiquitous, data on non-trade-policy barriers – which relate more broadly to the functioning of trade-facilitating markets such as transportation, insurance and distribution – are recorded only qualitatively. Second, in response, the literature has tried to measure the effects of non-trade-policy barriers by examining their effect on trade flows (e.g. Anderson and Neary, 2003; Baier, Bergstrand, and Feng, 2014; Dhingra, Freeman, and Huang, 2023). Although changes in trade barriers may indeed be reflected in changes in trade flows, trade flows are also shaped by other equilibrium effects induced by the policy, such as changes in global sourcing strategies, making it difficult to structurally interpret these estimates. Finally, unlike tariffs which are taxes and whose effects are well understood, non-trade-policy barriers operate through trade-facilitating markets, making it much harder to pinpoint how they affect allocations and welfare.

Yet, better understanding non-trade-policy barriers remains highly relevant. For example, although the Trade and Cooperation Agreement (TCA), which governs the post-Brexit trade relations between the EU and the UK, is essentially tariff- and quota-free, EU-UK trade fell considerably in 2022 (Freeman et al., 2024). This paper fills this gap by estimating the extent to which trade barriers rose following the UK's exit from the EU Single market (ESM). To address the concerns outline above, we measure changes in non-trade-policy barriers by estimating how the marginal cost of serving the UK market evolved relative to unaffected markets, and we quantify the importance of one key barrier: a sharp increase in freight rates for road transport following the introduction of the TCA.

We focus our analysis on the bottled water industry for two reasons. First, as mentioned before, estimating whether the TCA increased trade barriers requires controlling for potential endogenous firm responses to mitigate the impact of the policy change, such as relocating production to the UK (e.g., Flaaen, Hortaçsu, and Tintelnot, 2020). Because bottled water is differentiated by its specific source, relocating production is not possible. This ensures that the relative adjustment in prices and

marginal costs cannot be driven by production relocation. Second, bottled water provides an ideal setting to disentangle the drivers of non-trade-policy barriers. On the one hand, the UK's Most Favored Nations (MFN) tariff on bottled water is zero which rules out trade policy barriers related to rules-of-origin provisions and trade policy uncertainty in the run-up to the UK's exit from the ESM. On the other hand, bottled water has a simple value chain. After extraction, the water is bottled close to the source, shipped by truck and locally distributed. This allows us to credibly control for alternative cost factors and evaluate the importance of one particular trade barrier, i.e., the increase in freight rates.

We begin by providing reduced-form evidence that UK consumer prices of bottled water increased substantially after the introduction of the TCA. To this end, we build a panel of product-level consumer prices from household-level scanner datasets in Belgium, France, and the UK and compare the evolution of UK prices of imported bottled water to those in Belgium and France. Using a difference-in-differences estimator, we show that consumer prices in the UK closely tracked prices in Belgium and France before the introduction of the TCA, but diverged sharply afterward. In our preferred specification, we find that consumer prices of bottled water are 18% higher in the UK at the end of 2022 relative to Belgium and France. Importantly, a placebo experiment comparing prices in Belgium and French prices results in a precisely estimated zero effect.

We link the increase in UK consumer prices of imported bottled water to a sharp rise in freight rates for road transport. As no matched dataset of product-level consumer prices and freight costs exists, we proxy for incurred freight rates as best as possible. More specifically, we collect route-level freight rates between each main water source in our dataset and the forty most populous areas, for example from Evian-les-Bains in France, where Evian is bottled, to Birmingham in the UK. To trace how freight rates on UK-bound routes changed following the introduction of the TCA, we adopt a similar difference-in-differences model as before and compare the evolution of freight rates on UK-bound routes to those on routes bound for Belgium or France. We find that two years after the introduction of the TCA, freight rates on UK-bound routes increased by roughly €850 relative to routes with non-UK destinations. Like before, a placebo experiment comparing freight rates on routes with a Belgian destination to those with a French destination confirms that the effect is entirely driven by the increase in freight rates on UK-bound routes.

The reduced-form evidence is, however, only suggestive of increased non-trade-policy barriers following the implementation of the TCA for two reasons. First, in structural terms, non-trade-

policy barriers should be identified from changes in the destination-specific cost of serving a market rather than from changes in destination-specific prices. While production relocation is impossible in the bottled water industry, observed price changes may still be contaminated by firms' markup adjustment. Second, the difference-in-differences estimates only establish a qualitative relationship between changes in final consumer prices and freight rates. To quantify the contribution of freight rates to the change in non-trade-policy barriers, we require an estimate of how the cost of supplying the UK market would have evolved absent the increase in freight rates.

To make progress, we develop a partial equilibrium model of the bottled water industry in the final part of the paper. Consumer preferences for bottled water are represented by a static discrete choice framework in which consumers select a variety from the set of available varieties. Supplying bottled water varieties entails destination-specific marginal costs that depend on local distribution costs and production costs and trade costs, the latter capturing how transport costs and other non-trade-policy barriers may have changed with the TCA. Finally, equilibrium consumer prices are determined in a simultaneous-move Nash-Bertrand pricing game among manufacturers. The model addresses the aforementioned issues as follows. First, combining data on prices and quantities with firm's first-order conditions, we back out destination-specific marginal costs which we can leverage to estimate how non-trade-policy barriers changed following the TCA. Second, after estimating a flexible cost function for supplying bottled water to different markets, we simulate how the marginal cost of serving the UK would have changed had freight rates on UK-bound routes evolved like those in Belgium and France. Comparing this counterfactual path to the observed one allows us to quantify the contribution of higher freight rates to the increase in non-trade-policy barriers.

The equilibrium model yields the two insights. First, we estimate that non-trade-policy barriers rose by 17% following the TCA. Comparing this to the reduced-form evidence for consumer prices implies that exporters' markup adjustment barely limited pass-through into final consumer prices. Second, the marginal costs of supplying the UK market would have risen by only 11.6% had freight rates on UK-bound routes evolved similarly to freight rates in Belgium and France. Therefore, we conclude that the change in freight rates explains a little under 30% of the increase in non-trade-policy barriers associated with the TCA. Given that trucking services are complementary to a wide set of traded goods, these results suggest that the impact on the road transport industry was likely an important driver of the post-TCA increase in consumer prices and non-trade-policy barriers.

Related literature This paper contributes to three different strands of literature. First, it adds to the growing literature on deep trade integration. As trade-policy barriers between most high-income countries have reached historical lows, countries have increasingly adopted agreements that target behind-the-border barriers, such as product standards (Chen and Novy, 2011; Fontagné et al., 2015), regulation (Martincus, Carballo, and Graziano, 2015), and barriers in trade-facilitating markets. Much of this literature has focused on the reduced-form relationship between trade flows and participation in deep trade agreements (e.g. Baier, Bergstrand, and Feng, 2014; Dhingra, Freeman, and Huang, 2023), leaving us with a lack of structurally interpretable estimates of non-trade-policy barriers, as advocated for by Anderson and Neary (2003), and an understanding of which policies especially give rise to non-trade-policy barriers. This paper make progress on both fronts. Exploiting the institutional context of the bottled water industry, we estimate that the TCA increased non-trade-policy barriers by 17%. Furthermore, we show that the TCA disrupted a key trade-facilitating market – the market for international road transport – which accounts for a little under 30% of the increase in non-trade-policy barriers.

Second, this paper relates to a growing literature that models trade costs beyond the traditional iceberg formulation. Recent work has explored the implications of the round-trip effect in transportation (e.g., Behrens and Picard, 2011; Ishikawa and Tarui, 2018; Hayakawa, Ishikawa, and Tarui, 2020; Wong, 2022) and the additive nature of transport costs (Irrarrazabal, Moxnes, and Opromolla, 2015). Of particular relevance to is the work studying the endogenous determination of freight rates, either as a choice between technologies (e.g., Hummels and Schaur, 2013; Pascali, 2017), as arising from a routing problem (Ganapati, Wong, and Ziv, 2024), or as the choice variable in imperfectly competitive markets (Asturias, 2020; Brancaccio, Kalouptsidi, and Papageorgiou, 2020). This paper provides new causal evidence that endogenous adjustment in the road freight industry is a key mechanism through which deep trade agreements non-trade-policy barriers and market intergration more broadly.

Third, this paper contributes to the growing literature that studies Brexit as a rare case of economic disintegration (Sampson, 2017; Dhingra and Sampson, 2022). The unexpected referendum outcome has been leveraged to study exchange rate pass-through (Fernandes and Winters, 2021; Breinlich et al., 2022; Corsetti, Crowley, and Han, 2022), as well as the effect of news and uncertainty on stock market returns Breinlich et al. (2018), Davies and Studnicka (2018), and Fisman and Zitzewitz (2019) and on the real economy (Born et al., 2019; Broadbent et al., 2024; Hassan et al., 2024). Related work

by Steinberg (2019), Graziano, Handley, and Limão (2021), and Ahmad et al. (2023) study the effect of trade policy uncertainty – measured from the potential MFN tariffs that would apply absent an agreement – on trade flows. We contribute to this literature in two ways. First, whereas most papers focus on the period before the introduction of the TCA, like Bakker et al. (2022) and Freeman et al. (2024), we focus on the period after the implementation of the TCA. By exploiting detailed price variation and a structural model, we provide the first estimate of non-trade-policy barriers induced by the TCA. Second, we quantify the contribution of one particular non-trade-policy barrier – the endogenous change in freight rates for road transport – and find that it accounts for 30% of the rise in non-trade-policy barriers.

Finally, methodologically, this paper connects to the literature that exploits detailed spatial price dispersion to infer the level of trade barriers and assess market integration. For instance, Shiue and Keller (2007), Donaldson (2018), and Chatterjee (2023) use historical commodity price data to understand the level of market integration in Western Europe, China, and India, respectively. Whereas this literature predominantly examines within-country price dispersion to study domestic integration, this paper, like Bradford (2003), studies international market integration. However, unlike Bradford (2003), which focuses on cross-sectional price differences, we leverage variation over time and across destinations to estimate how trade costs changed following Britain’s disintegration from the EU.

The remainder of the paper is structured as follows. Section 2 describes the construction of the datasets and provides descriptive evidence indicative of the trade barriers induced by the TCA. Section 3 presents the reduced-form evidence that corroborates our claim that the TCA led to higher consumer prices and road freight rates in the UK. Section 4 outlines the structural model, and Section 5 reports the estimated structural parameters and quantitative results. Section 6 concludes.

2 Data and motivational evidence

In this section, we introduce the scanner data and freight rate data that form the basis of our dataset on product-level consumer prices and quantities of bottled water and route-level freight rates. We discuss the timeline of Brexit and highlight that prices and freight rates in the UK closely tracked their Belgian and French counterparts before the introduction of the TCA. In contrast, from January 2021 onwards, average UK consumer prices for bottled water and freight rates on UK-bound routes sharply increased relative to those in Belgium and France.

2.1 Data sources

2.1.1 Scanner data

We construct bottled water consumption prices from household-level scanner data in Belgium, France, and the UK. In each country, the data are gathered by a market research firm that equips households with a scanning device to record the product's barcode, the store in which the product was bought, the number of units they bought and the tax-inclusive amount expressed in local currency units (LCU). By combining information from barcode descriptions with data on the number of units, volume sold, and the expenditure, we compute quantity consumed in liter and consumer prices per liter. Our period of analysis is from January 2010 to 2022 and covers over 7 million transactions involving bottled water (Table 1).

Water varieties and firms. We define a bottled water variety as the interaction of a brand, its water source, whether the water is flavored or not, the type of packaging (glass, plastic, or other), and the package size in milliliters. For example, a 500ml plastic bottle of unflavored Evian water constitutes one variety. We now elaborate further on the product definition.

First, we include both branded and private-label water, as private-label water typically accounts for around half of total bottled water consumption by volume. To account for multi-product and multi-brand firms, we associate each variety with a firm identifier by matching information obtained from GS1 at the barcode level.¹

Second, we aggregate over two dimensions of horizontal differentiation. First, we do not define varieties at the level of particular flavors, but only track whether the variety is flavored or unflavored. As shown in Table A.4, which reports the unadjusted R^2 of regressions of final consumer prices on various fixed effects, a dummy capturing whether the variety is flavored or not already captures most of the variation in prices along this dimension. Second, Table A.4 also indicates that the carbon dioxide content of the water does not explain much variation in consumer prices conditional on the other product characteristics. For this reason, we also do not distinguish between sparkling and still waters.

Finally, the bottle size and package type are key dimensions of differentiation. For instance, Table A.3 shows that, conditional on other product characteristics, medium-sized bottles and large

¹GS1 is a consortium that coordinates the allocation of barcodes between firms, ensuring that barcodes identify unique products.

Table 1: Sample overview

Variable	Overall	BEL	FRA	UK
Nr. Households				
· NARTD	53,122	5,303	17,915	29,904
· Water	40,219	4,662	16,326	19,231
Transactions (· 1,000)	7,319	1,014	3,293	3,012
Nr. Products	660	297	265	214
Average consumer price				
· (LCU)	-	0.37	0.28	0.33
· (EUR)	-	0.37	0.28	0.40
Inside good share (rel. NARTD)				
· Expenditure	-	0.22	0.30	0.13
Nr. Production locations				
· All	122	71	67	48
· Local	-	10	56	35
Origin (volume share)				
· Belgium	-	0.211	0.001	0.000
· France	-	0.473	0.902	0.075
· UK	-	0.000	0.000	0.743
· Other	-	0.103	0.066	0.021
· Unknown	0.091	0.213	0.031	0.161

Notes: This table provides summary statistics for the sample that removed purchases of fringe products and in stores that account for few purchases. We provide summary statistics for the overall sample and for country separately. The number of products is obtained by counting the number of products defined as the interaction of a brand, a source, whether the product is flavored, the bottle size and the package type. When we have been able to retrieve the source, we can further distinguish between local and foreign varieties. The number of production locations is the count of the number of distinct production locations we were able to identify. The number of households is computed as the average number of distinct households that bought either bottled water (“water”) or within the Non-Alcoholic Ready-To-Drink (NARTD) categories, which includes bottled water, soft drinks and juices, in a given year. For the importance of bottled water, the value share is the average expenditure share of bottled water in the NARTD industry, computed based on all households (“uncond.”) or based on households that purchased water in that year (“cond.”). The inside good share is computed as the share in the population accounted for by households that purchased water at least once in a given quarter. In doing so, we weight households at their population weight. Prices are computed using population weights interacted with purchased volume. The share accounted for by different origins is computed as the share in overall purchased volume sourced from that particular origin.

bottles are respectively 40% and 80% cheaper per liter compared to small bottles. At the same time, Table A.4 illustrates that dummies for small ($\leq 750ml$), medium ($750ml, 1500ml$) and large bottles ($\geq 1500ml$) are not sufficient to capture a significant share of the variation in final consumer prices. As Table A.3 shows, bottled water sold in glass bottles is 25% more expensive than water sold in plastic bottles; we therefore incorporate package type as an additional source of differentiation.

Throughout the paper, we include varieties in the sample if they satisfy at least one of the following conditions. First, we include all varieties with a volume-based market share of at least 0.1%. We also include varieties that are sold in at least two national markets if they appear in at least 10 quar-

ters in either market. Table A.2 shows that the final sample covers 95% of the total quantity sold and more than 92% of the total number of transactions.

Production locations. We complement the scanner data with product-level information on production locations. This information is key to estimating whether trade barriers changed following the enactment of the TCA. This is because international trade is dominated by multinational firms (Antràs et al., 2024) that may serve markets as pure exporters (Melitz, 2003), through local affiliates (Helpman, Melitz, and Yeaple, 2004), or via export platforms (Tintelnot, 2016). Without information about production locations, it is impossible to disentangle changes in trade barriers from other changes in destination-specific costs stemming from, for instance, production relocation.

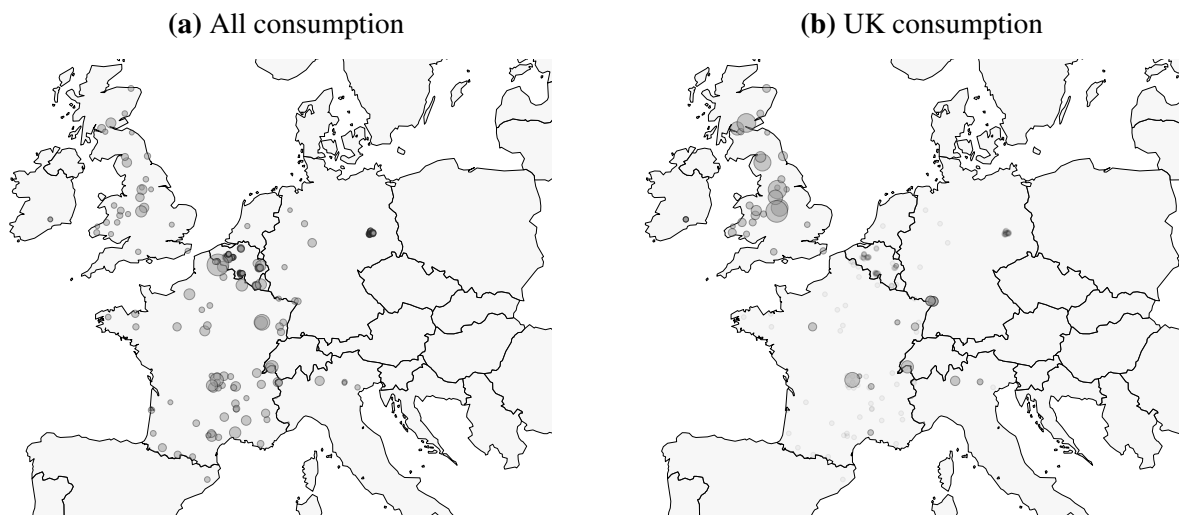
To this end, we rely on two institutional features of the bottled water industry. First, Directive 2009/54/EC regulates the branding and distribution of bottled water in the European Single Market (ESM) and requires manufacturers to disclose the water source from which the water is drawn on the bottle’s package. Second, to ensure consistency with the mineral and sanitary information disclosed to national health authorities and to avoid the need for specialized shipping, water is typically bottled close to its source. Importantly, although the UK left the ESM, it still abides by this regulation.

Based on these pieces of information, we hand-collect production locations. Table 1 shows that we identify the production location for over 90% of the total volume sold. When we are unable to determine the production location, this is typically because the variety is private-label.² Figure 1 plots the resulting spatial distribution of production locations. Panel A shows the distribution when pooling across all destinations, with the dots weighted by the total volume across all locations. Panel B instead weights locations by the volume consumed by UK households. This highlights that varieties imported into the UK are predominantly sourced from France and Italy.

Countries. The dataset comprises three European countries. In addition to the UK, we also include Belgium and France for two reasons. First, Table 1 shows that roughly 10% of the total volume purchased by UK households originates from abroad, with roughly 75% of this originating from France. Since water brands exported to the UK – such as Evian, Volvic and Vittel – are also sold domestically, we can construct a quasi-natural experiment comparing the evolution of UK consumer prices to those in France. Second, Table 1 shows that a little over 43% of bottled water sold in

²Unlike A-level brands, which use their particular source as a source of differentiation, private-label varieties tend to be more homogeneous and low-cost (see Table A.3). This gives retail chains more flexibility in terms of choosing from where to source their water, making it sometimes impossible to pinpoint the production location.

Figure 1: Production locations



Belgium also originates from France. Moreover, a considerable share of the bottled water varieties exported to the UK are also exported to Belgium. Including data from Belgium, therefore, allows us to conduct a placebo exercise where and compare how prices between a country pair that no longer trades under the ESM framework – the UK and France – evolved differently from a country pair that remained part of the ESM, i.e. Belgium and France.

Stores. The data cover purchases ranging from hypermarkets to small convenience stores. We restrict attention to retail chains with a volume-based market share above 0.1% and aggregate over different store formats operated by the same retail chain. Table A.1 shows that this restriction is relatively innocuous, as the main sample still covers 97% of the total volume sold.

2.1.2 Freight rates

A key contribution of this paper is to relate the change in consumer prices to the change in freight rates across destinations. Although the scanner data are rich in many dimensions, they do not record the transport costs that were incurred to deliver bottled water to various markets. For this reason, we turn to freight data from Upply, a private company that provides benchmarking and matching services for carriers and shippers.

To proxy transport costs as closely as possible, we adopt the following strategy. First, like most food items, bottled water is almost exclusively transported by truck. Since bottles are atmospherically sealed, they do not require special transportation. For this reason, we focus on freight rates for fully

loaded tautliner trucks that move general cargo and food.³ Second, we consider routes that depart in ZIP codes in which water production takes place and terminate in populous ZIP codes. We restrict the sample to routes whose origins and destinations jointly cover over 90% of bottle water consumption. The route from Evian-les-Bains, France to Leeds, United Kingdom is one such route. Finally, as shippers typically employ third-party carriers to cover shipments on the margin, we focus on spot prices that include fuel surcharges. This choice is innocuous, as contracted freight rates moved very similarly to the spot price around the introduction of the TCA. In this way, we obtain route-level freight rates at the monthly level from 2018, the startup year of Uply, until 2023.

2.1.3 Other data sources

We augment the consumption and the freight data with publicly available data. We obtain data on input prices of glass, plastic, and metal packaging from the Office of National Statistics (ONS) and the Institut National de La Statistique et des Etudes (INSEE). Similarly, we use data on wages in the manufacturing and retail sector from the ONS and INSEE. Finally, we gather data on the exchange rate between the British Pound (GBP) relative to the Euro (EUR) from the European Central Bank.

2.2 Brexit and the Trade and Cooperation Agreement

Timeline. Following a surprising and narrow victory for the Leave vote on 23 June 2016, the UK decided to leave the EU. The precise reasons why the vote tilted in favor of leaving the EU remain unclear (Becker, Fetzer, and Novy, 2017), but austerity-inspired welfare reforms following the financial crisis (Fetzer, 2019) and global competitive pressures on UK labor markets (Colantone and Stanig, 2018) presumably interacted with weak socio-economic fundamentals (Goodwin and Heath, 2016; Becker, Fetzer, and Novy, 2017).

The Leave vote was subsequently followed by a period marked by uncertainty about the UK's future stance on relations with the EU. Both so-called "hard" and "soft" Brexit scenarios were openly discussed, each with their respective projected effects (e.g. Reenen (2016), Dhingra et al. (2017), and Head and Mayer (2019)). However, on 23 January 2020, the European Union Withdrawal Agreement Act 2020 received Royal Assent and on 31 January 2020 the UK entered a transition period before finally leaving the EU on 31 December 2020.

By agreeing on the Trade and Cooperation Agreement (TCA), which governs current EU-UK

³Tautliner trucks are trucks to which trailers are attached that have sides made from plastic curtains or solid materials.

trade relations, the EU and UK effectively implemented a “soft” Brexit. Although the TCA stipulates tariff- and quota-free trade subject to rules-of-origin provisions, there are two reasons why transport between the EU and the UK may have become more cumbersome. On the one hand, time-consuming customs check were re-introduced, leading to long waiting lines at the EU-UK border. On the other hand, the agreement restricts market access for service providers and the free movement of persons. In particular, the qualifications of EU truck drivers are no longer automatically recognized in the UK (and vice versa) and service provision by foreign truck drivers under cabotage rules has become more stringent. Under the current market access provisions for trucking services, foreign truck drivers can make only two trips in the UK before having to return to the EU.

Consumer prices. Before turning to the formal reduced-form evidence, we first visually inspect whether consumer prices of bottled water in the UK diverged from their Belgian and French counterparts. To this end, we average over individual transactions and plot the aggregate evolution of final consumers of bottled water for each destination at the monthly frequency, expressed in local currency units in Figure 2. The period from the Brexit referendum to the inception of the TCA is shaded in grey, and key dates are marked with vertical dashed lines. We make three observations.

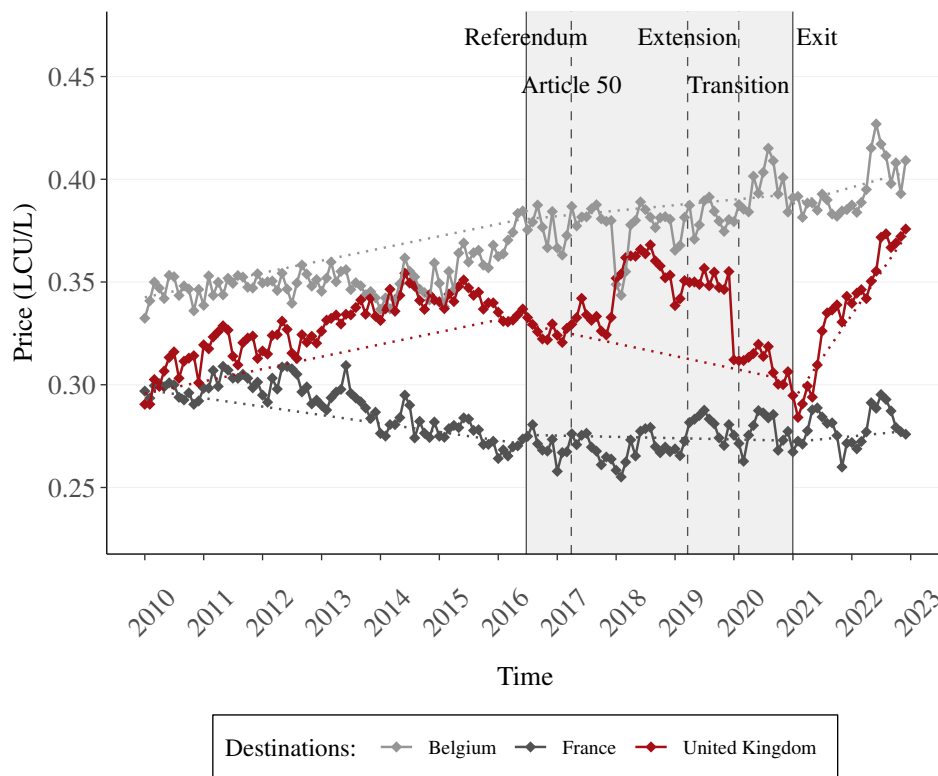
First, whereas UK consumer prices were on an upward trend prior to the referendum, they trended downward in the period between the Leave Vote and the UK’s withdrawal from the EU. This implies that aggregate UK consumer prices for bottled water, expressed in local currency, did not respond on impact to the sharp depreciation of the GBP relative to the EUR. As shown in Figure A.1, where we display the price evolution for foreign and domestic varieties separately, this pattern arises not only because prices for local varieties did not respond, but also because UK prices of foreign varieties did not change markedly after the depreciation. When we express UK final consumer prices in EUR in Figure A.2, we find that UK consumer prices in EUR sharply fell. Taken together, this suggests that bottled water prices are likely set in local currency, and we maintain this assumption throughout the analysis.

Second, not only did UK consumer prices fail to respond to the Leave vote on impact, they also continued on a downward trend relative to prices in Belgium and France. Figure A.1a demonstrates that this is because prices for local varieties were trending downward and because UK prices for imported varieties did not deviate from their gentle pre-depreciation trend (see Figure A.1b). Whereas other work has stressed the role of trade policy uncertainty in pushing up UK import prices during the intermediate period (e.g. Graziano, Handley, and Limão, 2021; Ahmad et al., 2023), we find a

very limited role for such uncertainty in the bottled water industry, consistent with UK MFN import tariffs on bottled water from the EU being zero.

Finally, Figure 2 shows that, in contrast to Belgium and France, consumer prices for bottled water sharply rose when the TCA came into force. Figure 2 reveals that UK consumer prices, on average, rose from £0.29 per liter to roughly £0.38 per liter. Moreover, not only did UK prices for imported water rise sharply (see Figure A.1b), but so did prices for local varieties (see Figure A.1a). In contrast, Belgian and French consumer prices were very stable when the TCA was introduced. Hence, while we can detect a clear break evolution in UK consumer prices, we are unable to do so for Belgium and France.

Figure 2: Evolution of consumer prices (LCU)



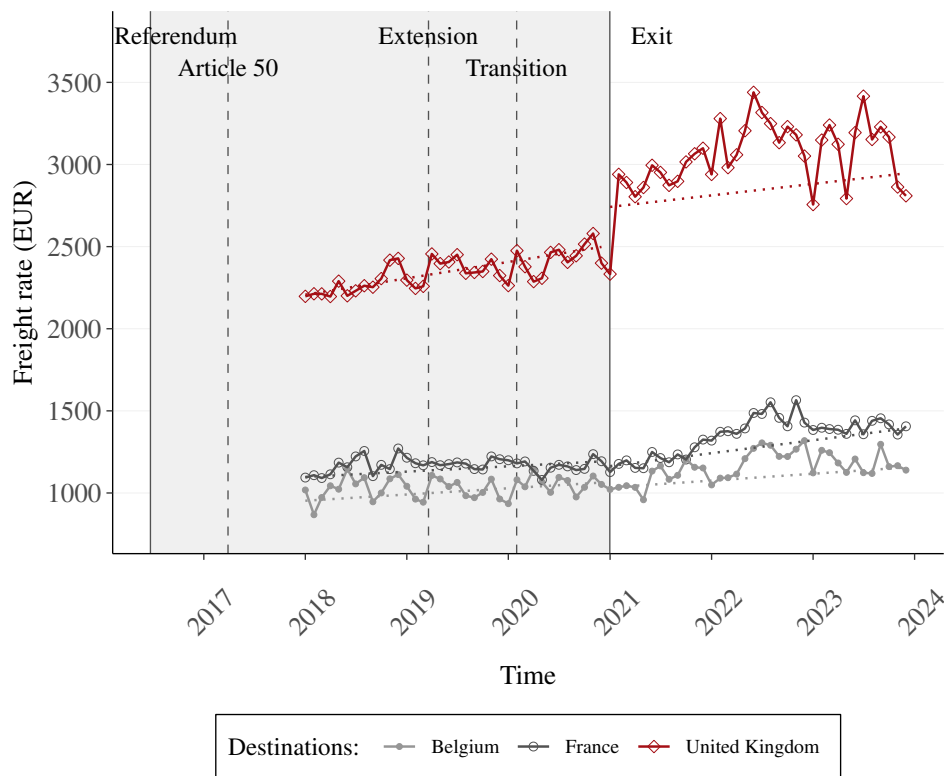
Notes: This figure plots the aggregate evolution of consumer prices for bottled water across Belgium, France, and the UK expressed in local currency units at a monthly frequency. We compute these series by aggregating over households, stores and bottled water varieties by weighting transaction-level unit prices using the interaction of population weights and volumes.

Freight rates. To more closely link the strong increase in consumer prices of bottled water to the introduction of the TCA, we next turn to the evolution of freight rates for road transport between Belgium, France, and the UK. Figure 3 plots the evolution of average freight rates for destinations Belgium, France and the UK separately. Like before, the period from 2018, which marks the start of

the freight data, to the UK's exit from the ESM is shaded in grey, and key dates are indicated with vertical dashed lines.

Before the UK exited the ESM, freight rates for routes with Belgian and French destinations were on a relatively flat trajectory, while those with UK destinations were trending slightly upward. However, freight rates for routes with UK destinations abruptly rose when the TCA came into force. Figure 3 indicates that these rates increased from roughly €2,300 to a little over €3,000 – an increase of about 30% on average increase. Although freight rates for routes with non-UK destinations began trending upward somewhat from 2022 onwards, they did not experience a discrete jump in January 2021, but rather mirrored the stability of consumer prices in Belgium and France for bottled water.

Figure 3: Evolution of freight rates



Notes: This figure shows the evolution of the average freight rates for Belgium, France and the UK separately at the monthly frequency. We associate freight rates with a given country when the destination of the route is in that specific country. We aggregate over routes by taking a simple average.

3 Effect of the TCA on prices and freight rates

While the aggregate evolution of UK consumer prices and freight rates on UK-bound routes relative to those in Belgium and France is suggestive of changing trade costs, these patterns could also be

driven by, for instance, compositional changes in the set of available varieties. This section presents evidence that the increase in UK prices and freight rates on UK-bound routes is causally linked to the introduction of the TCA.

3.1 Effect of the TCA on consumer prices

To substantiate this claim, we estimate the following difference-in-differences model:

$$\ln(p_{i,dt}) = \sum_{k \neq 2019} \beta_k \mathbb{1}[d = UK] \times \mathbb{1}[t = k] + \mathbf{X}'_{i,dt} \boldsymbol{\Gamma} + \varepsilon_{i,dt}. \quad (1)$$

Here, $p_{i,dt}$ denotes the final consumer price of variety i , sold in destination d at time t . We include treatment dummies by interacting two indicator functions: one $\mathbb{1}[d = UK]$ is one for observations in the UK, the other, $\mathbb{1}[t = k]$, equals one whenever the time period t coincides with period k . In this way, the coefficients of interest in Equation (1), β_k , measure the extent to which UK prices evolved differently before and after 2019 relative to those in Belgium and France.⁴ The vector $\mathbf{X}_{i,dt}$ contains various fixed effects, the exact specification of which is discussed below.

We estimate each specification using a weighted OLS estimator where the weights are the interaction between population weights and the volume sold. While the population weights are used to ensure the price evolution is representative, the volume is used to weight observations according to their importance in consumption. The results are, however, qualitatively similar when we consider population weights as weights or we do not weigh at all (see Table B.1 and B.2 respectively). We cluster standard errors at the variety level to account for correlation in $\varepsilon_{i,dt}$ over time and across destinations within a given variety.

Baseline results. Columns (1)-(3) of Table present the results when we include all varieties in the sample. Column (1) reports the results when we include variety and time fixed effects in the vector $\mathbf{X}_{i,dt}$ and $\mathbb{1}[d = UK]$ as this indicator is not yet subsumed by the fixed effects. In this way, we control for the cross-sectional heterogeneity in consumer prices across varieties – for instance due to persistent quality or cost differences – as well as the average evolution of consumer prices for bottled water across destinations. Before 2019, UK consumer prices were falling relative to Belgian and French consumer prices, which is consistent with the aggregate evolution of UK consumer prices

⁴Given that the sample period for the price data starts well before 2019, we pool all estimates before 2016 into one estimate.

shown in Figure 2. In 2020, the year preceding the introduction of the TCA, UK consumer prices were, on average, no longer higher compared to their Belgian and French counterparts. From 2021 onwards, however, UK consumer prices diverged significantly from Belgian and French consumer prices as they were 6.7% and 16.4% higher in 2021 and 2022.

Columns (2) and (3) consider the robustness of these results by including more detailed fixed effects. First, in addition to time fixed effects, column (2) now also includes variety-destination fixed effects, which subsume $\mathbb{1}[d = UK]$. In this specification, we recover quantitatively similar results. UK consumer prices were on a downward trajectory relative to prices in Belgium and France before 2019 and were 6% and 15.4% higher in 2021 and 2022, respectively. Second, column (3) retains the variety-destination fixed effects but replaces the time fixed effects with much more detailed variety-time fixed effects. These fixed effects fully control for the evolution of variety-level consumer prices that is common between destinations, for example due to changes in production costs, ensuring that the treatment effect is purely identified from variation between destinations within varieties. In this specification, we recover a stronger response: UK consumer prices rose by 13.6% and 17.6% in 2021 and 2022, respectively, relative to consumer prices in Belgium and France. Moreover, the variety-time fixed effects fully account for the earlier downward trend in UK prices relative to the other countries, leaving only post-2020 treatment effects statistically significant.

If the price response was indeed driven by the introduction of the TCA, one would expect prices of imported varieties to have increased in particular and immediately after the agreement came into force. First, in column (4) we restrict the sample to varieties that are imported into the UK. This ensures that the treatment effects are identified purely from variation in consumer prices between destinations for varieties that are exported to the UK and sold domestically in France and Belgium. Consistent with the previous results, before 2020, UK consumer prices of imported varieties closely tracked those in Belgium and France. However, from 2021 onwards, UK consumer prices of foreign varieties rose significantly by 12% and 18.8% in 2021 and 2022, respectively.

Second, Figure 4 presents the results when we estimate Equation (4) at the monthly level, including only varieties imported into the UK and controlling for variety-destination and variety-time fixed effects. In this case, we estimate the treatment effects relative to December 2019. In line with the results in column (4), we find that UK consumer prices of imported varieties were closely tracking their Belgian and French counterparts before 2021. From the beginning of 2021 onwards, however, UK consumer prices began to diverge and became statistically significantly higher from

mid-2021 onwards, ultimately stabilizing at a permanently higher level—between 15% and 20%.

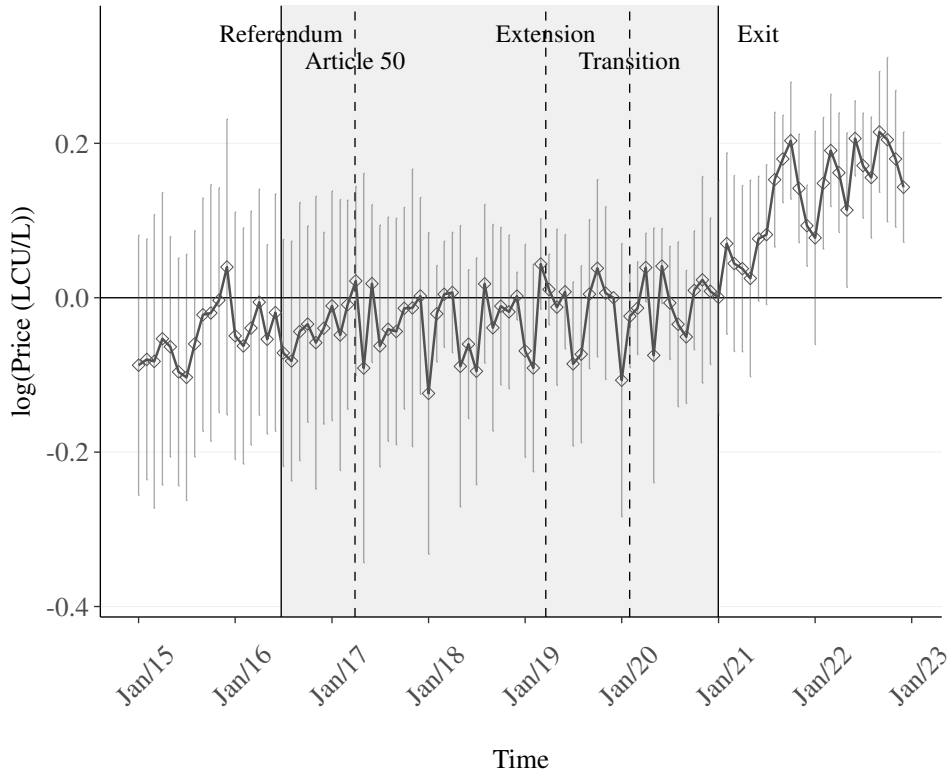
Table 2: Difference-in-differences - Consumer prices

	All			Foreign	Placebo	Domestic DiD	Local
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$\mathbb{1}[d = UK] \times \mathbb{1}[t \leq 2015]$	0.0431* (0.0261)	0.0344 (0.0261)	-0.0328 (0.0496)	-0.05 (0.0534)	-0.0547* (0.0322)	-0.0603 (0.0516)	0.0484 (0.0334)
$\mathbb{1}[d = UK] \times \mathbb{1}[t = 2016]$	0.0604*** (0.0198)	0.0529*** (0.0198)	-0.0139 (0.0308)	-0.0175 (0.0376)	0.0151 (0.0116)	-0.0661* (0.0364)	0.0774*** (0.0246)
$\mathbb{1}[d = UK] \times \mathbb{1}[t = 2017]$	0.0473*** (0.0139)	0.0425*** (0.0135)	-0.0063 (0.0287)	-0.00318 (0.0366)	0.00917 (0.012)	-0.0413* (0.023)	0.0628*** (0.0168)
$\mathbb{1}[d = UK] \times \mathbb{1}[t = 2018]$	0.0338** (0.0136)	0.0318** (0.0135)	-0.0214 (0.018)	-0.0171 (0.0211)	0.0491** (0.0201)	-0.00692 (0.0105)	0.0356*** (0.0104)
$\mathbb{1}[d = UK] \times \mathbb{1}[t = 2019]$	-	-	-	-	-	-	-
$\mathbb{1}[d = UK] \times \mathbb{1}[t = 2020]$	-0.0132 (0.0144)	-0.0117 (0.0141)	0.00145 (0.0119)	0.00848 (0.0171)	0.062*** (0.0185)	0.0131 (0.024)	-0.0151 (0.0156)
$\mathbb{1}[d = UK] \times \mathbb{1}[t = 2021]$	0.0672*** (0.0206)	0.0596*** (0.0206)	0.136*** (0.0211)	0.12*** (0.0207)	0.0355 (0.0323)	0.0409 (0.0297)	0.0431* (0.0237)
$\mathbb{1}[d = UK] \times \mathbb{1}[t = 2022]$	0.164*** (0.0256)	0.154*** (0.0256)	0.176*** (0.0247)	0.188*** (0.0289)	0.0119 (0.0162)	0.0128 (0.0344)	0.138*** (0.0294)
$\mathbb{1}[d = UK]$	-0.213*** (0.0725)	-	-	-	-	-	-
θ_i	✓						
$\theta_{i,d}$		✓	✓	✓	✓	✓	✓
λ_t	✓	✓				✓	✓
$\lambda_{i,t}$			✓	✓	✓		
R^2	0.009	0.009	0.005	0.064	0.300	0.012	0.017
No. obs	7,355	7,355	7,355	1,104	786	1,605	3,776

Notes: This table shows the results from estimating Equation (1) for different sets of controls and definitions of the treatment group using data at the yearly frequency. Columns (1)-(3) include both locally produced UK and varieties imported into the UK. In addition, column (1) includes variety and time fixed effects, column (2) includes variety-destination and time fixed effects and column (3) includes variety-destination and variety-time fixed effects. Column (4) includes variety-destination and variety-time fixed effects but focuses exclusively on varieties imported into the UK. Column (5) considers the same sample as column (4) but excludes UK consumer prices and considers a placebo experiment by defining the treatment group as consumer prices in Belgium. Columns (6) and (7) both include variety-destination and time fixed effects but differ in the sample and the definition of the treatment group. Column (6) only includes UK consumer prices on locally produced UK and varieties imported into the UK and defines the treatment group as UK consumer prices of imported varieties. Column (7) only includes consumer prices of locally sourced varieties in Belgium, France and the UK and defines the treatment group as consumer prices of locally sourced UK varieties. We estimate each specification using a weighted OLS estimator where the weights are the interaction between population weights and the volume sold. We cluster standard errors at the variety level and denote statistical significance at the $p < 0.1^*$, $p < 0.05^{**}$ or $p < 0.01^{***}$ level.

Alternative estimates. We now turn to a set of alternative specifications. The first two provide supporting evidence that the previously reported estimates have a causal interpretation, while the final two compare our approach – exploiting variation between destination markets – to an alternative approach that uses variation between product origins. The latter comparison highlights that not only UK prices of imported products but also those of domestic varieties responded strongly to the introduction of the TCA.

Figure 4: Difference-in-differences: Consumer prices - Foreign products



Notes: This figure plots the results when we estimate Equation (1) using data at the monthly level, we only include consumer prices of varieties that are imported into the UK and when we define the treatment group to be UK consumer prices. In this case, we estimate the treatment effects relative to December 2019. Like before, we weigh observations using population weights interacted with purchased volumes. We cluster standard errors at the variety level and report 95% confidence intervals along the point estimates.

First, to assess the internal validity of our approach, column (5) reports the results from a placebo experiment. In this specification, we retain only Belgian and French consumer prices for varieties imported into the UK but counterfactually assign Belgium as the “treated” destination. This allows us to test whether Belgian consumer prices diverged from French consumer prices after 2021. Although Belgian prices were slightly higher in 2018 and 2020 relative to 2019, column (5) shows that Belgian and French consumer prices of varieties imported into the UK did not diverge after 2021. Since the introduction of the TCA should have affected only UK consumer prices, finding no comparable price increase in Belgium provides additional support that the observed rise in UK consumer prices was indeed caused by the inception of the TCA.

Second, in addition to a difference-in-differences model, we also considered a triple difference-in-differences model. In this case, we limit the control group to only include French consumer prices and examine how Belgian and UK consumer prices jointly (first treatment group) and UK prices alone (second treatment group) evolved relative to French consumer prices. Figure B.1 shows the results

of this approach using data at a monthly frequency, once including all varieties (Figure B.1a) and once including only varieties imported into the UK (Figure B.1b). Reassuringly, Figures B.1a and B.1b shows that the treatment effects for the joint Belgian-UK group are largely insignificant, while the treatment effects for the second group are very much in line with the difference-in-difference estimates presented in Table and Figure 4.

Third, how do the results from our approach, which exploits price variation between destinations, compare to an approach that relies on price variation across products within a given destination? For instance, to understand the effect of Trump tariffs, Flaaen, Hortaçsu, and Tintelnot (2020) compare the price evolution of washing machines to the evolution for ranges and Fajgelbaum et al. (2020) exploit variation between products targeted with tariffs and those that were not. Such an approach uncovers the true effect of the policy change as long as untargeted products are not indirectly affected, for example through industry or general equilibrium effects. To implement this approach, in Column (6), we restrict attention to UK consumer prices and compare how prices of imported varieties changed relative to local one. Before 2021, there is weak evidence that UK consumer prices of foreign varieties were rising relative to those of locally sourced varieties. However, after 2021, we find a precisely estimated null effect. Therefore, we cannot reject the null hypothesis that UK consumer prices of foreign and local varieties changed similarly after 2021.

Finally, column (6) suggests that a substantial increase in the price local varieties may have also been an important driver of rising UK prices. To illustrate this point more clearly, we confine the sample to Belgian, French and UK consumer prices of locally produced and consumed varieties. This allows us to compare how UK consumer prices of locally produced varieties evolved relative to prices of locally produced varieties produced in Belgium and France. Although this specification only allows for variety and time fixed effects, it still provides insight into how local prices evolved across destinations. Column (7) shows that UK consumer prices of locally produced varieties were on a downward trajectory prior to 2019. However, in line with the results from column (6), UK consumer prices of locally produced varieties increased significantly relative to those in Belgium and France after 2020. As we cannot include variety-time fixed effects, we are unable to determine whether this increase due to industry equilibrium effects such as reduced competitive pressure as in (Crowley, Han, and Prayer, 2024; Amiti and Heise, 2024) or due to general equilibrium effects that affected input prices as suggested by Born et al. (2019) and Broadbent et al. (2024). Nevertheless, these results the importance of leveraging variation between destinations rather than across products

within the UK to estimate the effects of the UK's secession on consumer prices.

3.2 Effect of the TCA on freight rates

Next, we provide causal evidence that a part of the increase in UK prices for bottled water was due to a sudden increase in freight rates induced by the TCA. To this end, we estimate a difference-in-difference model similar to the one in the previous section:

$$t_{od,t} = \sum_{k \neq 2019} \beta_k \mathbb{1}[d = UK] \times \mathbb{1}[t = k] + \mathbf{X}'_{od,t} \boldsymbol{\Gamma} + \varepsilon_{od,t}. \quad (2)$$

Here, $t_{od,t}$ is the freight rate on a route with origin o and destination d at time t . As before, the indicator function, $\mathbb{1}[d = UK]$, is one for observations in the UK, and $\mathbb{1}[t = k]$, equals one whenever the time period t coincides with period k . In this way, the coefficients of interest, β_k , measure the extent to which freight rates on UK-bound routes evolved differently before and after 2019 relative to those with destinations in Belgium and France. The vector $\mathbf{X}_{od,t}$ contains various fixed effects, the exact specification of which is discussed below.

In contrast to the estimation for consumer prices, we estimate how the *level* of freight rates on routes with UK destinations changed relative to routes with non-UK destinations.⁵ Table 3.1 reports the results from estimating Equation (2) for different sets of fixed effects using data at the yearly frequency and an unweighted OLS estimator. We cluster standard errors at the route level to account for serial correlation in $\varepsilon_{i,dt}$ over time.

Baseline results. Columns (1)-(3) present the results when we include all routes in the sample. Column (1) includes year fixed effects, and destination fixed effects, which subsume the treatment indicator. The inclusion of destination fixed effects controls for persistent differences in freight rates between destinations, for instance due to their remoteness or local demand for transportation services. Year fixed effects control for changes in freight rates that are shared across routes, such as continent-wide changes in fuel prices. We find that in 2018, freight rates on routes with UK destinations were on average €79 cheaper relative to 2019 and €56 more expensive in 2020 compared to routes with non-UK destinations. This mild pre-TCA pre-trend, however, was dwarfed by the sharp post-TCA

⁵We base this decision on the fact that there is substantial literature that argues that freight rates are best modeled as additive trade costs (Hummels and Skiba, 2004; Hummels, Lugovskyy, and Skiba, 2009; Irarrazabal, Moxnes, and Oromolla, 2015; Ishikawa and Tarui, 2018). For this reason, examining how freight rates changed in absolute levels is more suitable.

increase in freight rates: UK-bound routes were €486, €585 and €581 higher in 2021, 2022 and 2023, respectively, more expensive compared to routes with non-UK destinations.

Columns (2) and (3) assess the robustness of these results by adding more detailed fixed effects. Column (2) replaces the destination fixed effects with route-level fixed effects to fully account for persistent differences in freight rates at the route level. This specification yields quantitatively similar results to those in column (1). In Column (3), we further account for changes in freight rates over time that are shared among routes with the same origin. As the treatment effects are now identified solely using variation in freight rates between destinations within a given route, this is our preferred specification. While the results before the TCA are relatively similar to the ones in column (1) and (2), freight rates on UK-bound routes respond much more strongly in this specification, increasing by €756, €907 and €862 more than those on routes with a non-UK destination in 2021, 2022 and 2023, respectively.

As argued previously, if the increase in freight rates was indeed driven by the TCA, the effects in columns (1) - (3) should be driven by routes originating outside the UK, and the timing of the effect should coincide with the introduction of the TCA. First, column (4) constrains the sample to routes with non-UK origins, ensuring that the treatment effects are estimated solely from changes in freight rates between destinations that share the same non-UK origins. Indeed, column (4) confirms that the majority of the effect reported in column (3) is driven by increases in freight rates on routes entering the UK.

Second, Figure 5 plots the results from estimating the specification underlying column (4) at the monthly frequency, with effects normalized to December 2019. Consistent with earlier results, there is a mild upward trend in freight rates on routes with UK destinations relative to those with non-UK destinations prior to 2021. However, when the TCA came into force in January 2021, freight rates on UK-bound routes rose sharply, almost immediately settling at a level roughly €1,000 higher than before.

Alternative estimates. We now provide additional evidence supporting a causal interpretation of the results presented so far. In addition, we examine whether freight rates on routes with UK origins and destinations changed as well.

First, a causal interpretation of the results requires that the control group – routes with non-UK destinations – was not affected by the introduction of the TCA. Because the TCA did not alter trade relations between Belgium and France, we test whether freight rates on routes with Belgian

Table 3: Difference-in-differences - Freight rates

	All			Foreign	Placebo	Domestic DiD	Local
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$\mathbb{1}[d = UK] \times \mathbb{1}[t = 2018]$	-79.7*** (4.69)	-79.7*** (4.66)	-114*** (4.79)	-117*** (4.78)	28.6*** (3.09)	-97.4*** (5.74)	-21*** (2.3)
$\mathbb{1}[d = UK] \times \mathbb{1}[t = 2019]$	-	-	-	-	-	-	-
$\mathbb{1}[d = UK] \times \mathbb{1}[t = 2020]$	56.2*** (4.91)	56.2*** (4.88)	97.8*** (3.82)	102*** (3.5)	56*** (2.78)	116*** (4.7)	-0.914 (1.99)
$\mathbb{1}[d = UK] \times \mathbb{1}[t = 2021]$	486*** (22.8)	486*** (22.7)	756*** (15.2)	773*** (14.5)	66.2*** (4.67)	698*** (18.1)	110*** (5.96)
$\mathbb{1}[d = UK] \times \mathbb{1}[t = 2022]$	585*** (36.5)	585*** (36.2)	907*** (24.3)	915*** (24.4)	-34*** (12.7)	864*** (40.3)	156*** (13.3)
$\mathbb{1}[d = UK] \times \mathbb{1}[t = 2023]$	551*** (37.3)	547*** (33)	862*** (22.6)	867*** (22.6)	-56.8*** (12.1)	818*** (35.3)	124*** (12.7)
θ_d	✓						
θ_{od}		✓	✓	✓	✓	✓	✓
λ_t	✓	✓				✓	✓
$\lambda_{o,t}$			✓	✓	✓		
R^2	0.013	0.324	0.787	0.809	0.131	0.551	0.306
No. obs	3,756	3,756	3,756	2,681	1,648	1,814	2,008

Notes: This table shows the results from estimating Equation (2) for different sets of controls and definitions of the treatment group using data at the yearly frequency. Columns (1)-(3) include freight rates on routes with domestic UK origins and foreign origins. In addition, column (1) includes destination and time fixed effects, column (2) includes origin-destination and time fixed effects and column (3) includes origin-destination and origin-time fixed effects. Column (4) includes origin-destination and origin-time fixed effects but confines the sample to freight rates on routes with non-UK origins. Column (5) considers the same sample as column (5) but excludes freight rates with UK destinations and considers a placebo experiment by defining the treatment group as freight rates on routes with a destination in Belgium. Columns (6) and (7) both include origin-destination and time fixed effects but differ in the sample and the definition of the treatment group. Column (6) only includes freight rates on routes with UK destinations and defines the treatment group as freight routes on routes with a non-UK origin. Column (7) only includes freight rates on local routes, i.e. routes with an origin and destination in the same country and defines the treatment group as freight rates on UK domestic routes. We estimate each specification using an unweighted OLS estimator, cluster standard errors at the route level and denote statistical significance at the $p < 0.1^*$, $p < 0.05^{**}$ or $p < 0.01^{***}$ level.

destinations changed differently from those with French destinations. Column (5) confirms that freight rates on routes with Belgian destinations remained closely aligned with those with French destinations before and after 2021. This evidence, combined with the relatively flat profile of average freight rates on routes with Belgian and French destinations shown in Figure 3, strengthens the case that the sudden increase in freight rates documented in columns (1)-(4) was indeed caused by the UK's secession from the EU.

Second, to address the mild upward pre-trend in freight rates with UK destinations relative to non-UK destinations, we estimate a triple difference-in-differences model. The control group consists of routes with French destinations, the first treatment group includes routes with Belgian and UK destinations and the second treatment group are solely routes with UK destinations. In this way, we estimate the differential change in freight rates on routes with UK destinations while

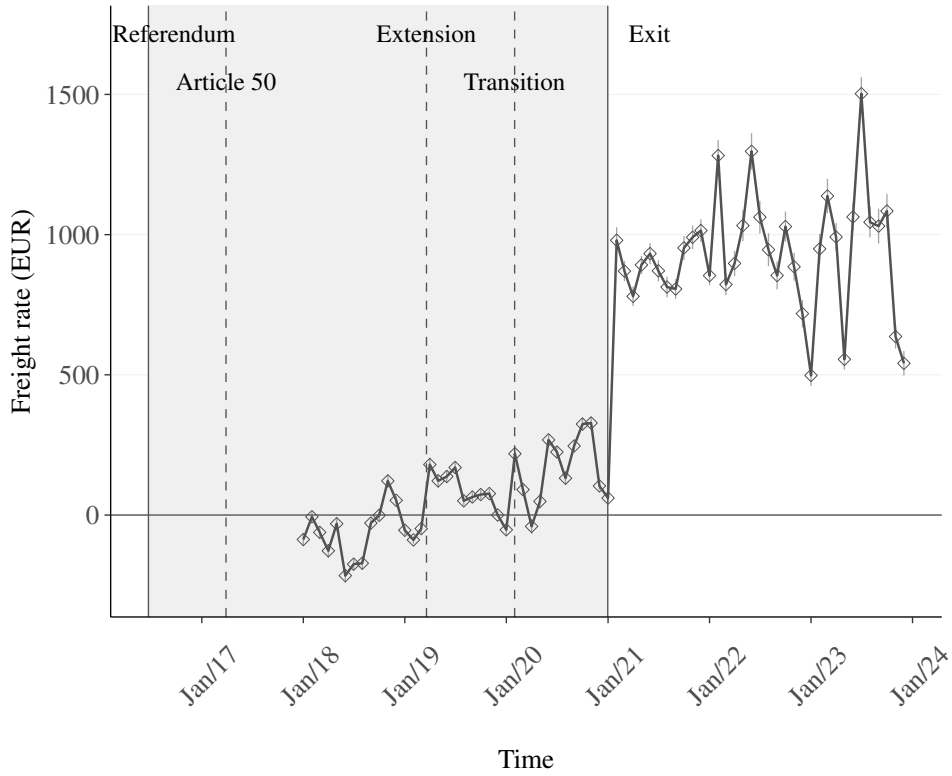
allowing a common evolution in freight rates on Belgian and UK routes relative to those with French destinations. Figures B.3a present the results for the full sample and Figure B.3b does so for the subsample of routes with non-UK origins. While the large and sudden response after 2021 remains unaffected, we estimate a much weaker pre-trend trend the second treatment group. This suggests that the mild pre-trend did not an underlying upward trend in UK-bound freight rates, but rather a slight downward trend in those with French destinations.

Third, we study whether freight rates on routes with UK origins and destinations, i.e. domestic routes, increased after 2021 as well. Column (7) shows the results from estimating Equation (2) when we limit the sample to routes with UK destinations and compare how freight rates on routes with non-UK origins changed relative to those with UK origins. We find very similar results compared to column (5): freight rates on routes with UK origins were, on average, €800 more expensive than with UK origins in 2023. This suggests that, unlike consumer prices, freight rates on routes with non-UK origins and UK destinations diverged substantially from those with UK origins and destinations.

Were freight rates with UK origins and destinations not affected at all by the TCA? In column (8), we restrict the sample to routes domestic routes and we compare how freight rates on domestic UK routes changed relative to domestic Belgian and French routes. We find that freight rates on domestic UK routes rose, depending on the horizon, by €110 and €150 relative to those on domestic routes in France and Belgium. Hence, consistent with the framework of Brancaccio, Kalouptsi, and Papageorgiou (2020), we find evidence of equilibrium spillovers from the market for international road transportation to the market for domestic routes.

Taking stock. This section has established that UK consumer prices for bottled water increased substantially after the introduction of the TCA, driven by higher prices for both locally sourced and imported varieties. Moreover, a key contributing factor appears to be a sudden and large increase in freight rates on routes with UK destinations. This evidence is, however, only suggestive of heightened non-trade-policy barriers following the TCA, since non-trade-policy barriers should be identified from variation in the costs of serving a market rather than from changes in prices. Furthermore, so far, our analysis has only established a qualitative link between rising freight rates and consumer prices. To address these issues, we turn to a structural model of supply and demand in the bottled water industry next.

Figure 5: Difference-in-difference - Freight rates



Notes: This figure plots the results when we estimate Equation (2) using data at the monthly level, we only include freight rates on routes with non-UK origins and when we define the treatment group to be freight rates on routes with UK destinations. In this case, we estimate the treatment effects relative to December 2019. The results are estimated using an unweighted OLS estimator, we cluster standard errors at the route level and report 95% confidence intervals along the point estimates.

4 Structural model

4.1 Preferences for bottled water

Consumers make a static discrete choice by maximizing random utility over bottled water varieties available in their market. We adopt this framework for three reasons. First, Table 1 shows that the share of consumers purchasing bottled water varies substantially across countries, likely due to differences in the perceived quality of tap water. A discrete choice model naturally accommodates an outside option to account for the fact that not all consumers purchase bottled water.

Second, we focus on a static rather than a dynamic choice. Because bottled water is storable, ignoring inventory decisions might bias the estimated substitution patterns Hendel and Nevo (2006).⁶ To mitigate this concern, we aggregate the data to the quarterly level, as roughly 90% of the

⁶Ignoring such dynamic concerns typically leads to an overestimation of demand elasticities and an underestimation of the degree to which manufacturers will want to raise prices above marginal costs.

households buy bottled water each quarter (Table 1).⁷

Finally, consumers only choose among varieties available in their regional market. Although there are no formal checks along, for instance, the Belgium–France border and consumers are free to engage in cross-border shopping,⁸ we abstract from this behavior for two reasons. On the one hand, bottled water is inexpensive, voluminous, and heavy, making spatial arbitrage unprofitable. On the other hand, cross-border shopping is marginal in the data: such transactions account for less than 2% of all purchases in Belgium and France.

Formally, consumer i in market t – where a market is defined as the combination of a quarter in a location – chooses a variety j from the set of available varieties $\mathcal{J}_t$ to maximize their indirect utility $V_{ij,t}$, which is given by:

$$V_{ij,t} = \alpha \ln(P_{j,t}) + \mathbf{X}_{j,t}' \boldsymbol{\beta} + \lambda_t + \xi_{j,t} + \zeta_{g(j),t} + \tilde{\varepsilon}_{ij,t}.$$

Here, $P_{j,t}$ is the price of a variety j in market t . Prices enter logarithmically, which implies that expenditure on bottled water is a constant fraction of total expenditures and price elasticities are quasi-constant in prices (Björnerstedt and Verboven, 2016).⁹ The vector $\mathbf{X}_{j,t}$ contains observable product characteristics, and λ_t are market-level fixed effects that capture variation in the importance of the outside good across markets. Following the literature, we normalize the indirect utility of the outside option to zero in all markets. The term $\xi_{j,t}$ is an unobserved demand shifter which is allowed to be correlated with prices. Finally, while $\zeta_{g(j),t}$ is a random effect that accounts for correlation in the preferences of consumers across groups, branded and private label water in the application, $\varepsilon_{ij,t}$ captures idiosyncratic tastes for products that are individually and independently distributed across products and markets. Following the literature, we assume that $\zeta_{g(j),t} + \varepsilon_{ij,t}$ are distributed such that

⁷This statistic is based on the countries where bottled water is appealing relative to the outside option.

⁸Auer et al. (2023) study how the incidence of Swiss franc appreciation differs between consumers living close to and far from national borders.

⁹Given the large variation in prices per liter, having prices enter in levels would imply counterfactually large variation in own-price elasticities.

the model-implied market shares, $\sigma_{j,t}$, take the following form:¹⁰

$$\sigma_{j,t} = \frac{\exp(\delta_{j,t}/(1-\sigma)) \exp(I_{g,t})}{\exp(I_{g,t}/(1-\sigma)) \exp(I_t)} \quad (3)$$

where $\delta_{j,t} \equiv \alpha \ln(P_{j,t}) + \mathbf{X}'_{j,t} \boldsymbol{\beta} + \lambda_t + \xi_{j,t}$ is the mean utility of variety j in market t , $I_{g,t} \equiv (1-\sigma) \ln \left(\sum_{j \in \mathcal{J}_t^g} \exp(\delta_{j,t}/(1-\sigma)) \right)$ is the inclusive value of selecting a variety from group $g \in \mathcal{G}$ and $I_t \equiv \ln \left(\sum_{g \in \mathcal{G}} \exp(I_{g,t}) \right)$. In this model, substitution between goods is mainly governed by two parameters. On the one hand, the price-sensitivity parameter α governs substitution away from goods that become more expensive. On the other hand, when the correlation in preferences across varieties belonging to the same group – which is governed by σ – is strong, substitution will mostly be directed towards goods of the same group.

4.2 Market structure and marginal costs

Consumer prices are determined as the equilibrium of a simultaneous move Bertrand game in which manufacturers choose prices to maximize profits. Formally, they are the solution to the following system of first-order conditions:

$$\mathbf{p}_t = -(\boldsymbol{\Delta}_t \odot \boldsymbol{\Omega}_t)^{-1} \cdot \boldsymbol{\sigma}_t(\mathbf{p}_t; \boldsymbol{\Theta}^d) + \mathbf{c}_t \quad (4)$$

In this expression, $(\boldsymbol{\Delta}_t \odot \boldsymbol{\Omega}_t)^{-1} \cdot \boldsymbol{\sigma}_t(\mathbf{p}_t; \boldsymbol{\Theta}^d)$ is the vector of markups charged by manufacturers. This depends on (1) the matrix of first-order derivatives of demand with respect to prices $\boldsymbol{\Delta}_t$, (2) the ownership matrix $\boldsymbol{\Omega}_t$ which internalizes substitution towards other varieties sold by the same manufacturer, (3) and the vector of market shares $\boldsymbol{\sigma}_t(\mathbf{p}_t; \boldsymbol{\Theta}^d)$, which depends on the demand parameters $\boldsymbol{\Theta}^d$. The vector $\mathbf{c}_t$ denotes destination-specific marginal costs.

Armed with demand estimates and the first-order conditions, Equation 4 allows us to back out destination-specific marginal costs. To quantify the contribution of changes in freight rates to changes in non-trade-policy-barriers barriers, we must parameterize the destination-specific marginal cost

¹⁰The particular functional form assumption we make is the following:

$$\zeta_{g(j)} + \tilde{\varepsilon}_{ij,t} \sim F(\varepsilon) = \exp \left\{ - \sum_{g \in \mathcal{G}} \left(\sum_{j \in \mathcal{J}_t^g} \exp[-\varepsilon/(1-\sigma)] \right) \right\}^{1-\sigma}.$$

function. For this reason, we assume that the cost function takes the following form:

$$c_{j,t} = \gamma \left(\frac{1}{k} \sum_k t_{s(j),dt-k} \right) + \beta W_t + \omega_{j,t} + \eta_{j,t} \quad (5)$$

Equation 5 accounts for three sources of destination-specific marginal costs. First, they may differ between destinations when local distribution costs W_t differ (Burstein, Eichenbaum, and Rebelo, 2005; Crucini, Telmer, and Zachariadis, 2005; Parsley and Wei, 2007). We capture those differences by including labor unit costs in the retail sector, expressed in LCU per hour from Eurostat. Second, as shown in section 3, freight rates on UK-bound routes increased substantially after the introduction of the TCA. Therefore, differences in destination-specific marginal costs may also arise from differences in transport costs $t_{s(j),t}$. As γ measures the share accounted for by freight rates in the total marginal cost of bottled water provision, it is a key parameter to determine the importance of freight rates in overall non-trade-policy-barriers. Finally, the cost function allows for two additional sources of variation: production costs, captured by $\omega_{j,t}$, and $\eta_{j,t}$, which is an unobserved variety- and market-specific cost shifter.

We highlight two features of this parameterization. First, by combining data on production locations with data on purchases of the same variety in multiple locations, we can fully and flexibly account for the marginal costs of production through variety-time fixed effects. Hence, our estimates of trade costs are valid under arbitrary returns to scale and scope. Second, because the cost function is additive in levels, we assume that the production function of bottled water takes a Leontief form. In the bottled water industry, these assumptions are natural: both transport and distribution are essential for delivering bottled water to the final consumer. Moreover, assuming that distribution and transport costs enter the cost function additively is standard in the international trade literature (Corsetti and Dedola, 2005; Parsley and Wei, 2007; Irarrazabal, Moxnes, and Opromolla, 2015).

5 Quantitative exercise

To evaluate the extent to which non-trade-policy barriers have changed and to quantify the contribution of rising freight rates, we proceed in two steps. First, we estimate the key demand and supply parameters of the model in a consistent manner. Second, we combine the estimated model with a plausible counterfactual series of freight rates that would have prevailed in the absence of the

TCA. This allows us to predict how marginal costs would have evolved under this counterfactual scenario and to compare the resulting series of marginal costs to those implied by the observed path of prices and market shares.

5.1 Estimates of demand and supply

We begin by estimating the demand parameters, focusing in particular on the price sensitivity parameter α and the parameter σ which governs the correlation of preferences within branded and private label water. Armed with these estimates, we then rely on equation (4) to recover marginal costs. In turn, allows us to estimate the contribution of freight rates for road transportation to destination-specific costs as measured by the parameter γ .

Demand estimates. To estimate the demand parameters, we return to Equation (3) and, following Berry (1994), invert the demand function to arrive at the following estimable equation:

$$\ln \left(\frac{s_{j,t}}{s_{0,t}} \right) = \alpha \ln (P_{j,t}) + \mathbf{X}'_{j,t} \boldsymbol{\beta} + \sigma \log (s_{j|g,t}) + \xi_{j,t}$$

Here, $s_{j,t}$ and $s_{0,t}$ denote the expenditure shares on variety j and on outside option, respectively, in market t , serving as the empirical counterparts to the model-implied market shares in Equation (3). Also, $s_{j|g,t}$ is the expenditure share of variety j computed within its group. This illustrates that the parameter σ is identified from the comovement between the overall and the within-group expenditure shares: the stronger the correlation in preferences within groups, the more substitution occurs among products of the same group, and hence the greater the role of the within-group share in determining total expenditure shares.

To estimate the demand parameters consistently, we need to overcome two endogeneity concerns. First, if unobserved demand shocks, $\xi_{j,t}$, are partly controlled by firms, they are likely correlated with prices $P_{j,t}$. For instance, if favorable weather conditions drive up the demand for certain varieties, firms may recognize this and increase prices. We address this by employing an international version of the well-known Hausman (1996)-instrument. Traditionally, Hausman (1996) instruments use variation in prices of the same product in other regions of the same country, $\frac{1}{|M|} \sum_{m \neq t} P_{j,t}$, as instruments for local prices, $P_{j,t}$, under the assumption that regional cost shocks are correlated while demand shocks are not. However, if demand shocks are correlated across regions – for instance due to national advertising campaigns – the exclusion restriction may fail. We

therefore construct international Hausman instruments that exploit variation in prices of the same product across national markets. Specifically, we instrument $P_{j,t}$ with $\frac{1}{|M|} \sum_{m \neq n(t)} P_{j,t}$. This approach has two advantages. On the one hand, since many bottled water varieties are produced in the same location, cross-country price variation predominantly reflects common cost shocks—such as input prices, energy costs, or transport costs—rather than demand variation. On the other hand, because most advertising and promotional activities are organized at the national level and cross-border shopping is negligible (below 2% of all transactions), it is much less likely that demand shocks are correlated across the relevant international markets.

Second, by construction, the within-group share $s_{j|g,t}$ is mechanically related to the overall share $s_{j,t}$ and is therefore correlated with the unobserved demand shock $\xi_{j,t}$. To obtain a consistent estimate of σ , we instrument $s_{j|g,t}$ with the number of available varieties within group g in market t . This variable is strongly correlated with the within-group share, as a larger number of products mechanically lowers $s_{j|g,t}$ through greater intra-group competition. At the same time, provided that firms determine their product portfolios prior to the realization of the market-specific demand shocks, the number of available products is plausibly exogenous to $\xi_{j,t}$ and thus serves as a valid instrument.

Panel (A) of Table 4 presents the results from estimating Equation (5.1). Column (1) shows the OLS results. With OLS, we recover negative but inelastic demand estimates, which is consistent with the presence of unobserved demand shocks driving both expenditure shares and prices. In addition, we estimate a near-perfect correlation between overall and within-group expenditure shares. This is also consistent with the mechanical correlation induced by the simultaneous determination of $s_{j,t}$ and $s_{j|g,t}$.

Column (2) turns to the 2SLS results, where we instrument prices with the international Hausman (1996)-instrument and the within-group shares with the number of products within each group g . Based on the first stage F-stats, both instruments appear sufficiently strong. Moreover, both 2SLS coefficients move in the expected direction relative to the OLS results. In particular, the estimated price sensitivity parameter α implies a median own price elasticity of -12.03, while σ – which governs the correlation in preferences across varieties belonging to the same group – is estimated at 0.86. This indicates a strong but not perfect correlation in preferences and substitution within groups of branded and private label varieties.

Supply estimates. With the demand estimates in hand, we next compute the substitution matrices Δ_t to back out the vector of destination-specific marginal costs using Equation (4). Given the relatively elastic own-price elasticities, model-implied markups are modest and negative marginal costs account only for a tiny share of the observations.

We estimate the extent to which freight rates for road transport contribute to destination-specific marginal cost using an OLS estimator for two reasons. First, because we observe consumption of the same variety across multiple markets, we can fully account for production costs at the variety-level by including variety-time fixed effects. Including these detailed fixed effects absorbs all unobserved costs shocks that may affect prices similarly across destinations and freight rates. Second, transporting bottled water does not require specialized transportation, making it plausible that producers take the prevailing freight rate as given.

Panel (B) of Table 4 reports the results from estimating Equation (5), once including only contemporaneous freight rates and once with an additional lag. Column (1) shows no contemporaneous relationship between freight rates and destination-specific marginal costs. This likely reflects the short-run price stickiness observed in consumer markets: while freight rates adjusted immediately following the introduction of the TCA (see Figure 5), consumer prices responded only significantly after roughly six months (see Figure 4). Consistent with this lagged adjustment, Column (2) shows a positive and statistically significant effect. The point estimate of 0.12 implies that a €1,000 increase in freight rates leads to a 12-cent-per-liter rise in destination-specific marginal costs.

5.2 The importance of freight rates as a non-trade policy barrier

To quantify the extent to which non-trade policy barriers increased and how the increase in freight rates contributed to this rise, we proceed in two steps. First, we construct the counterfactual series of freight rates that approximates how transport costs would have evolved in the absence of the TCA. Second, we compare the actual and counterfactual paths of destination-specific marginal costs to evaluate the impact of the observed increase in freight rates.

Counterfactual series of freight rates. We take an econometric approach to constructing a realistic counterfactual series of freight rates that would have prevailed in the absence of the TCA. First, we compile a sample that includes all freight rates on routes with non-UK origins and destinations, as well as freight rates on routes involving the UK up to December 2020—the month before the

Table 4: Demand and Supply Estimation

(A) Demand estimation	OLS	IV
$\ln(s_{j,dt}/s_{0,dt})$	(1)	(2)
$\log(p_{j,dt})$	-0.0287* (0.016)	-1.69** (0.772)
$\log(s_{j g,dt})$	1*** (0.003)	0.86*** (0.143)
N	28,275	28,275
1 st Stage F-stat : $\log(p_{j,dt})$	-	41.20
1 st Stage F-stat : $\log(s_{j g,dt})$	-	25.95
$\theta_{j,d}$	✓	✓
λ_{dt}	✓	✓
Median Own Price Elasticity	-	-12.03
Median Market Price Elasticity	-	-0.35
(B) Supply estimation	Contemp.	Lagged
$\hat{m}c_{j,dt}$	(1)	(2)
$\frac{1}{k} \sum_k t_{j,dt-k}$ (1'000 EUR)	-0.0122 (0.034)	0.124** (0.060)
N	28,853	28,853
Adj. R^2	0.71	0.72
$\omega_{j,t}$	✓	✓
W_t	✓	✓

Notes: This table shows the demand estimates in Panel (A) and the supply estimates in Panel (B). Panel (A) presents the OLS estimates in column (1) and column (2). The reported F-statistics are computed according to Montiel-Olea and Pflueger (2013) and valid under clustered standard errors. Panel (B) presents the supply estimates with only contemporaneous freight rates in column (1) and column (2) adds one lag as well. In both cases we include fixed effects at the variety-time level and retail wages in the destination market as controls. We cluster standard errors at the product-destination level and denote statistical significance at the $p < 0.1^*$, $p < 0.05^{**}$ or $p < 0.01^{***}$ level.

introduction of the TCA. Second, we estimate the following regression:

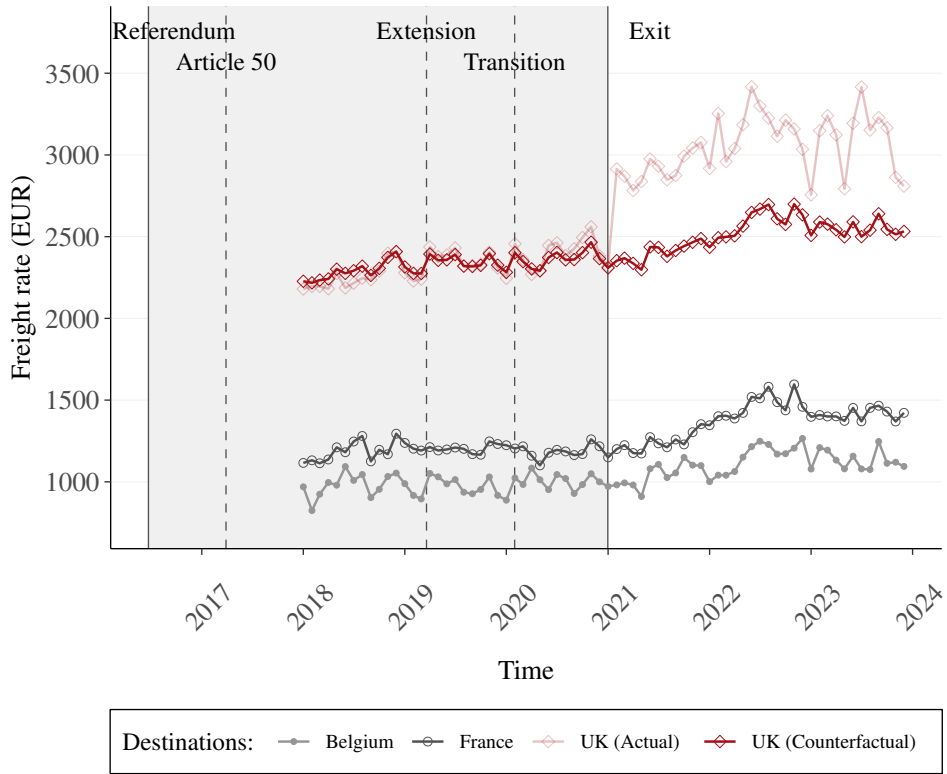
$$t_{od,t} = \theta_{od} + \lambda_t + \varepsilon_{od,t}. \quad (6)$$

Here, $t_{od,t}$ denotes the freight rate on a route with origin o and destination d at time t . The route-level fixed effects θ_{od} capture persistent differences in freight rates across routes, while the time fixed effects λ_t capture the evolution of freight rates common to all routes. Importantly, although the time fixed effects are estimated using data that include UK routes prior to the TCA, they rely only on freight rates for routes between Belgium and France after 2021. This ensures that the post-2021 evolution reflects conditions in EU markets unaffected by the TCA. Finally, we predict how freight rates with UK-origins and -destinations would have evolved had the TCA not occurred based on the

estimated fixed effects as follows: $t_{od,t}^c = \hat{\theta}_{od} + \hat{\lambda}_t$. In this way, we assume that freight rates on routes with UK destinations would have followed the same trajectory as those on routes bound for Belgium and France.

Similar to Figure 3, Figure 6 plots the evolution of average freight rates on routes with destinations in Belgium, France and the UK. In contrast to Figure 3, we now also overlay the series of actual freight rates on routes with UK destinations (light red) with the counterfactual series that shows how freight rates with UK destinations would have evolved in the absence the TCA (dark red). Before the introduction of the TCA, the actual and counterfactual series closely track each other. However, after 2021, the two series sharply diverge. Importantly, the counterfactual series also captures the rise in freight rates that would have occurred even without the TCA due to global supply chain disruptions in 2022.

Figure 6: Counterfactual freight rates



Notes: The figure plots the average actual evolution of freight rates with destinations in Belgium, France, and the UK. It also shows our prediction of how average freight rates on routes with UK-destinations would have evolved in the absence of the TCA. This prediction is based on taking the estimated fixed effects from Equation (6) and constructing the counterfactual series of freight rates as: $t_{o,d,t}^c = \hat{\theta}_{o,d} + \hat{\lambda}_t$.

TCA-induced non-trade policy barriers. Table 5 contains our main structural estimates. Each column presents results from estimating the same specification underlying column (4) of Table 3.1, but for different outcome variables, using data starting in 2018.¹¹ Column (1) presents the results from estimating a difference-in-differences model comparing how UK consumer prices of imported varieties changed relative to prices of the same varieties in Belgium and France. As this is the same specification as the one in column (4) of Table 3.1, albeit estimated on a slightly different sample, we estimate a very similar 17% relative increase in the prices of UK consumer prices of imported bottled water.

After recovering destination-specific marginal costs from Equation (4), column (2) turns to how destination-specific marginal costs of supplying bottled water to the UK changed relative to Belgium and France. The results indicate that the observed increase in consumer prices primarily reflects higher destination-specific marginal costs.¹² In line with consumer prices, the marginal costs of supplying bottled water to the UK market rose by 17% relative to the cost of supplying the same varieties to the Belgian and French markets. As we control for changes in production costs, changes in destination-specific marginal cost map directly into changes in trade barriers. We, therefore, conclude that the TCA caused an increase of 17% in non-trade-policy barriers.

Importance of changes in freight rates. To what extent was this increase in relative costs caused by higher freight rates? To answer this question, columns (3) and (4) present how marginal costs and consumer prices would have evolved in the UK relative to Belgium and France if freight rates had evolved according to the counterfactual path. In this counterfactual scenario, column (3) shows that the marginal cost of supplying bottled water to the UK would have only gone up by 11.6% after two years. We therefore predict that non-trade-policy barriers would have only gone up by 11.6%. Comparing this counterfactual change to the 17% increase implied by the structural model and the data in column (2), we attribute just under 30% of the rise in non-trade-policy barriers to higher road freight rates.

Column (4) turns to the implied effects on consumer prices. In the counterfactual scenario, consumer prices of bottled water in the UK would have increased by 12.3%, roughly 25% less than the actual observed increase of 17% reported in column (1). While these results suggest that rising

¹¹We trim the sample to 2018 as before 2018 we have no data on freight rates and therefore we can only compute counterfactual equilibria from 2018 onwards

¹²Given the relatively elastic demand estimates reported in Table 4, markups are small. Within the model, changes in relative prices thus mainly reflect changes in relative marginal costs.

Table 5: Quantifying Non-Trade Policy Barriers

y	$\ln(p_{jt})$	$\ln(\hat{m}c_{jt})$	$\ln(mc_{jt}^{\text{Counter}})$	$\ln(p_{jt}^{\text{Counter}})$
	(1)	(2)	(3)	(4)
$T_{i,d} \times \mathbb{1}(t = 2018)$	-0.0337** (0.0154)	-0.0339** (0.0154)	-0.0191 (0.0132)	-0.0189 (0.0165)
$T_{i,d} \times \mathbb{1}(t = 2019)$	-	-	-	-
$T_{i,d} \times \mathbb{1}(t = 2020)$	0.00812 (0.0114)	0.00645 (0.0114)	0.00449 (0.0118)	0.00567 (0.0121)
$T_{i,d} \times \mathbb{1}(t = 2021)$	0.109*** (0.0203)	0.108*** (0.0192)	0.0672*** (0.0193)	0.0871*** (0.0194)
$T_{i,d} \times \mathbb{1}(t = 2022)$	0.171*** (0.0358)	0.17*** (0.0336)	0.116*** (0.0271)	0.123*** (0.0282)
Median of LHS	0.91	0.80	0.79	0.88
$\theta_{i,d}$	✓	✓	✓	✓
$\lambda_{i,t}$	✓	✓	✓	✓
R^2	0.377	0.378	0.193	0.236
No. obs	378	378	378	378

Notes: This table shows the results from applying the difference-in-differences estimator from of column (4) Table 3.1 in which we compare the evolution in outcomes for varieties imported into the UK to outcomes of those same varieties in Belgium and France. In contrast to Table 3.1, we now restrict the sample to data starting in 2018 and aggregate to the quarterly level. Column (1) reports the results for consumer prices, column (2) for backed-out marginal costs. Columns (3) and (4) present the results when we apply this estimator to counterfactual marginal costs and consumer prices that would have occurred according to our structural model had the TCA not increased freight rates. We cluster standard errors at the product-destination level and denote statistical significance at the $p < 0.1^*$, $p < 0.05^{**}$ or $p < 0.01^{***}$ level.

freight rates played an important role in driving up costs and prices, the majority of the increase in non-trade-policy barriers following the TCA stemmed from other frictions—such as administrative delays, compliance costs, or other supply chain disruptions.

6 Conclusion

Although non-trade-policy barriers are essential to explain why international trade is much lower relative to domestic trade and often motivate deep trade agreements, their nature remains poorly understood. This paper make progress by studying the introduction of the Trade and Cooperation Agreement, which governs trade relations between the UK and the EU from 2021 onwards.

We focus on the bottled water industry, which is ideally suited to shed new light on the drivers of non-trade-policy barriers. This sector was not subject to tariff uncertainty prior to the TCA, is not subject to new tariffs or quotas under the TCA, and offers limited scope for production relocation. To conduct the analysis, we build a new dataset combining final product-level consumer prices and

route-level freight rates for road transport in the UK, Belgium and France.

We find that UK consumer prices rose by roughly 17% relative to those in Belgium and France following the introduction of the TCA. In addition, we document that freight rates for international road transport also rose by roughly 30% on UK-bound routes relative to routes bound for Belgium and France. After estimating a structural model of demand and supply in the bottled water industry, we estimate that non-trade-policy barriers rose by 17% of which roughly 30% can be attributed to the increase in freight rates.

While our analysis provides novel causal evidence on the magnitude of non-trade-policy barriers and their sources by relating them to one particular trade-facilitating market – the market for international road transport – we take the change in freight rates as given. An promising avenue for future work is to examine whether this rise in freight rates stemmed from physical constraints, such as longer travel times or less efficient capacity use in the international road transport market, or from institutional factors, such as more restrictive labor regulations governing who could serve EU-UK routes.

References

- Ahmad, S., N. Limão, S. Oliver, and S. Shikher. “Brexit Uncertainty and Its (Dis)Service Effects”. *American Economic Journal: Economic Policy* 15 (2023), 459–485. DOI: [10.1257/pol.20200808](https://doi.org/10.1257/pol.20200808).
- Amiti, M. and S. Heise. “US Market Concentration and Import Concentration”. *The Review of Economic Studies* (2024).
- Anderson, J. E. and P. J. Neary. “The mercantilist index of trade policy”. *International Economic Review* 44 (2003), 627–649. DOI: [10.1111/1468-2354.t01-1-00083](https://doi.org/10.1111/1468-2354.t01-1-00083).
- Antràs, P., E. Fadeev, T. C Fort, and F. Tintelnot. “Exporting, Global Sourcing and Multinational Activity: Theory and Evidence from the United States”. *Review of Economics and Statistics* Forthcoming (2024), 1–45.
- Asturias, J. “Endogenous Transportation Costs”. *European Economic Review* 123 (Apr. 2020). DOI: [10.1016/j.euroecorev.2019.103366](https://doi.org/10.1016/j.euroecorev.2019.103366).
- Auer, R., A. Burstein, S. Lein, and J. Vogel. “Unequal Expenditure Switching: Evidence from Switzerland”. *The Review of Economic Studies* Forthcoming (2023).
- Baier, Scott L., Jeffrey H. Bergstrand, and Michael Feng. “Economic Integration Agreements and the Margins of International Trade”. *Journal of International Economics* 93 (2014), 339–350. DOI: [10.1016/j.jinteco.2014.03.005](https://doi.org/10.1016/j.jinteco.2014.03.005).
- Bakker, J.D., N. Datta, R. Davies, and J. De Lyon. “Non-tariff Barriers and Consumer Prices - Evidence from Brexit”. *mimeo* (2022).
- Becker, S. O., T. Fetzner, and D. Novy. “Who Voted for Brexit? A Comprehensive District-level Analysis”. *Economic Policy* 32 (2017), 601–650. DOI: <https://doi.org/10.1093/epolic/eix017>.
- Behrens, K. and P. M. Picard. “Transportation, Freight Rates, and Economic Geography”. *Journal of International Economics* 85 (Nov. 2011), 280–291. DOI: [10.1016/j.jinteco.2011.06.003](https://doi.org/10.1016/j.jinteco.2011.06.003).
- Berry, Steven. “Estimating Discrete-Choice Models of Product Differentiation”. *RAND Journal of Economics* 25 (1994), 242–262.
- Björnerstedt, Jonas and Frank Verboven. “Does merger simulation work? Evidence from the Swedish analgesics market”. *American Economic Journal: Applied Economics* 8 (2016), 125–164. DOI: [10.1257/app.20130034](https://doi.org/10.1257/app.20130034).

- Born, B., G. J. Müller, M. Schularick, and P. Sedláček. “The Costs of Economic Nationalism: Evidence from the Brexit Experiment”. *The Economic Journal* 129 (Oct. 2019), 2722–2744. DOI: [10.1093/ej/uez020](https://doi.org/10.1093/ej/uez020).
- Bradford, Scott. “Paying the Price: Final Goods Protection in OECD Countries”. *The Review of Economics and Statistics* 89 (2003), 24–37. DOI: [10.1162/003465303762687686](https://doi.org/10.1162/003465303762687686).
- Brancaccio, G., M. Kalouptsi, and T. Papageorgiou. “Geography, Transportation, and Endogenous Trade Costs”. *Econometrica* 88 (2020), 657–691. DOI: [10.3982/ECTA15455](https://doi.org/10.3982/ECTA15455).
- Breinlich, H., E. Leromain, D. Novy, and T. Sampson. “The Brexit Vote, Inflation and UK Living Standards”. *International Economic Review* 63 (Feb. 2022), 63–93. DOI: [10.1111/iere.12541](https://doi.org/10.1111/iere.12541).
- Breinlich, H., E. Leromain, D. Novy, T. Sampson, and A. Usman. “The Economic Effects of Brexit: Evidence from the Stock Market”. *Fiscal Studies* 39 (Dec. 2018), 581–623. DOI: [10.1111/1475-5890.12175](https://doi.org/10.1111/1475-5890.12175).
- Broadbent, B., F. Di Pace, T. Drechsel, R. Harrison, and S. Tenreyro. “The Brexit Vote, Productivity Growth, and Macroeconomic Adjustments in the U.K.” *The Review of Economic Studies* (Sept. 2024). DOI: [10.1093/restud/rdad086](https://doi.org/10.1093/restud/rdad086).
- Burstein, A., M. Eichenbaum, and S. Rebelo. “Large Devaluations and the Real Exchange Rate”. *Journal of Political Economy* 113 (2005), 742–784.
- Chatterjee, S. “Market Power and Spatial Competition in Rural India”. *The Quarterly Journal of Economics* 138 (Aug. 2023), 1649–1711. DOI: [10.1093/qje/qjad004](https://doi.org/10.1093/qje/qjad004).
- Chen, N. and D. Novy. “Gravity, Trade Integration, and Heterogeneity across Industries”. *Journal of International Economics* 85 (Nov. 2011), 206–221. DOI: [10.1016/j.jinteco.2011.07.005](https://doi.org/10.1016/j.jinteco.2011.07.005).
- Colantone, I., G. Ottaviano, and P. Stanig. “The Backlash of Globalization”. *Handbook of International Economics* 5 (2022), 405–477. DOI: [10.1016/bs.hesint.2022.02.007](https://doi.org/10.1016/bs.hesint.2022.02.007).
- Colantone, I. and P. Stanig. “Global Competition and Brexit”. *American Political Science Review* 112 (May 2018), 201–218. DOI: [10.1017/S0003055417000685](https://doi.org/10.1017/S0003055417000685).
- Corsetti, G., M. Crowley, and L. Han. “Invoicing and the Dynamics of Pricing-to-market: Evidence from UK Export prices around the Brexit Referendum”. *Journal of International Economics* 135 (Mar. 2022). DOI: [10.1016/j.jinteco.2022.103570](https://doi.org/10.1016/j.jinteco.2022.103570).

- Corsetti, Giancarlo and Luca Dedola. “A macroeconomic model of international price discrimination”. *Journal of International Economics* 67 (Sept. 2005), 129–155.
- Crowley, M., L. Han, and T. Prayer. “The Pro-competitive Effects of Trade Agreements”. *Journal of International Economics* 150 (2024), 1–19. DOI: [10.1016/j.jinteco.2024.103936](https://doi.org/10.1016/j.jinteco.2024.103936).
- Crucini, M., C. Telmer, and M. Zachariadis. “Understanding European Real Exchange Rates”. *American Economic Review* 95 (2005), 724–738.
- Davies, R. B. and Z. Studnicka. “The Heterogeneous Impact of Brexit: Early Indications from the FTSE”. *European Economic Review* 110 (Nov. 2018), 1–17. DOI: [10.1016/j.euroecorev.2018.08.003](https://doi.org/10.1016/j.euroecorev.2018.08.003).
- Dhingra, S., R. Freeman, and H. Huang. “The Impact of Non-tariff Barriers on Trade and Welfare”. *Economica* 90 (Jan. 2023), 140–177. DOI: [10.1111/ecca.12450](https://doi.org/10.1111/ecca.12450).
- Dhingra, S., H. Huang, G. Ottaviano, P. Pessoa, T. Sampson, J. Van Reenen, G. Gancia, and U. Panizza. “The Costs and Benefits of Leaving the EU”. *Economic Policy* 32 (2017), 651–705. DOI: [10.2307/26566881](https://doi.org/10.2307/26566881).
- Dhingra, S. and T. Sampson. “Expecting Brexit”. *Annual Review of Economics* 14 (2022), 495–519. DOI: [10.1146/annurev-economics](https://doi.org/10.1146/annurev-economics).
- Donaldson, D. “Railroads of the Raj: Estimating the Impact of Transportation Infrastructure”. *American Economic Review* 108 (Apr. 2018), 899–934. DOI: <https://doi.org/10.1257/aer.20101199>.
- Fajgelbaum, P., P. Goldberg, P. Kennedy, and A. Khandelwal. “The Return to Protectionism”. *The Quarterly Journal of Economics* 135 (2020), 1–55. DOI: [10.1093/qje/qjz036](https://doi.org/10.1093/qje/qjz036).
- Fernandes, A. P. and A. L. Winters. “Exporters and Shocks: Impact of the Brexit Vote Shock on Bilateral Exports to the UK”. *Journal of International Economics* 131 (July 2021). DOI: [10.1016/j.jinteco.2021.103489](https://doi.org/10.1016/j.jinteco.2021.103489).
- Fetzer, T. “Did Austerity Cause Brexit?” *American Economic Review* 109 (2019), 3849–3886. DOI: [10.1257/aer.20181164](https://doi.org/10.1257/aer.20181164).
- Fisman, R. and E. Zitzewitz. “An Event Long-Short Index: Theory and Applications”. *American Economic Review: Insights* 1 (Dec. 2019), 357–372. DOI: [10.1257/aeri.20180399](https://doi.org/10.1257/aeri.20180399).
- Flaaen, A., A. Hortaçsu, and F. Tintelnot. “The Production Relocation and Price Effects of US Trade Policy: The Case of Washing Machines”. *American Economic Review* 110 (2020), 2103–2127. DOI: [10.1257/aer.20190611](https://doi.org/10.1257/aer.20190611).

- Fontagné, L., G. Orefice, R. Piermartini, and N. Rocha. “Product standards and margins of trade: Firm-level evidence”. *Journal of International Economics* 97 (Sept. 2015), 29–44. DOI: [10.1016/j.jinteco.2015.04.008](https://doi.org/10.1016/j.jinteco.2015.04.008).
- Freeman, R., K. Manova, T. Prayer, and T. Sampson. “Not to Belittle NTBs: Non-tariff Barriers and Trade during Brexit”. *mimeo* (2024).
- Ganapati, Sharat, Woan Foong Wong, and Oren Ziv. “Entrepôt: Hubs, Scale, and Trade Costs”. *American Economic Journal: Macroeconomics* 16 (2024), 239–278. DOI: [10.1257/mac.20220250](https://doi.org/10.1257/mac.20220250). URL: <https://pubs.aeaweb.org/doi/10.1257/mac.20220250>.
- Goldberg, P.K. and N. Pavcnik. “The Effects of Trade Policy”. *Handbook of Commercial Policy*. Vol. 1A. 2016, 161–206. DOI: [10.1016/bs.hescop.2016.04.002](https://doi.org/10.1016/bs.hescop.2016.04.002).
- Goodwin, M. J. and O. Heath. “The 2016 Referendum, Brexit and the Left Behind: An Aggregate-level Analysis of the Result”. *Political Quarterly* 87 (July 2016), 323–332. DOI: [10.1111/1467-923X.12285](https://doi.org/10.1111/1467-923X.12285).
- Graziano, A. G., K. Handley, and N. Limão. “Brexit Uncertainty and Trade Disintegration”. *The Economic Journal* 131 (Apr. 2021), 1150–1185. DOI: [10.1093/ej/ueaa113](https://doi.org/10.1093/ej/ueaa113).
- Hassan, T. A., S. Hollander, L. Van Lent, and A. Tahoun. “The Global Impact of Brexit Uncertainty”. *Journal of Finance* 79 (Feb. 2024), 413–458. DOI: [10.1111/jofi.13293](https://doi.org/10.1111/jofi.13293).
- Hausman, J. “Valuation of New Goods under Perfect and Imperfect Competition”. *The Economics of New Goods*. Ed. by Timothy F. Bresnahan and Robert J. Gordon. Vol. 58. 1996, 209–248. ISBN: 0226074153.
- Hayakawa, K., J. Ishikawa, and N. Tarui. “What Goes Around Comes Around: Export-enhancing Effects of Import-tariff reductions”. *Journal of International Economics* 126 (Sept. 2020). DOI: [10.1016/j.jinteco.2020.103362](https://doi.org/10.1016/j.jinteco.2020.103362).
- Head, K. and T. Mayer. “Gravity Equations: Workhorse, Toolkit, and Cookbook”. *Handbook of International Economics*. Vol. 4. Elsevier B.V., 2014, 131–195.
- Head, Keith and Thierry Mayer. “Brands in motion: How frictions shape multinational production†”. *American Economic Review* 109 (2019), 3073–3124. DOI: [10.1257/aer.20161345](https://doi.org/10.1257/aer.20161345).
- Helpman, E., M. Melitz, and S. Yeaple. “Export versus FDI with Heterogeneous Firms”. *American Economic Review* 94 (2004), 300–316. DOI: [10.1257/000282804322970814](https://doi.org/10.1257/000282804322970814).

- Hendel, Igal and Aviv Nevo. “Measuring the Implications of Sales and Consumer Inventory Behavior”. *Econometrica* 74 (2006), 1637–1673.
- Hummels, D., V. Lugovskyy, and A. Skiba. “The Trade-reducing Effects of Market Power in International Shipping”. *Journal of Development Economics* 89 (May 2009), 84–97. DOI: [10.1016/j.jdeveco.2008.05.001](https://doi.org/10.1016/j.jdeveco.2008.05.001).
- Hummels, D. and A. Skiba. “Shipping the Good Apples Out? An Empirical Confirmation of the Alchian-Allen Conjecture”. *Journal of Political Economy* 112 (2004), 1384–1402. DOI: [10.1086/422562](https://doi.org/10.1086/422562).
- Hummels, David L. and Georg Schaur. “Time as a Trade Barrier”. *American Economic Review* 103 (2013), 2935–2959.
- Irarrazabal, A., A. Moxnes, and L. D. Opromolla. “The Tip of the Iceberg: A Quantitative Framework for Estimating Trade Costs”. *Review of Economics and Statistics* 97 (2015), 777–792. DOI: [10.1162/REST_a_00517](https://doi.org/10.1162/REST_a_00517).
- Ishikawa, J. and N. Tarui. “Backfiring with Backhaul Problems: Trade and Industrial Policies with Endogenous Transport Costs”. *Journal of International Economics* 111 (Mar. 2018), 81–98. DOI: [10.1016/j.jinteco.2017.12.004](https://doi.org/10.1016/j.jinteco.2017.12.004).
- Martincus, C. Volpe, J. Carballo, and A.D. Graziano. “Customs”. *Journal of International Economics* 96 (May 2015), 119–137. DOI: [10.1016/j.jinteco.2015.01.011](https://doi.org/10.1016/j.jinteco.2015.01.011).
- Melitz, M. “The Impact of Trade on Intra-Industry Reallocations and Aggregate Industry Productivity”. *Econometrica* 71 (2003), 1695–1725. DOI: [10.1111/1468-0262.00467](https://doi.org/10.1111/1468-0262.00467).
- Montiel-Olea, J. and C. Pflueger. “A Robust Test for Weak Instruments”. *Journal of Business and Economic Statistics* 31 (July 2013), 358–369.
- Parsley, D. and S.-J. Wei. “A Prism into the PPP Puzzles: The Micro-Foundations of Big Mac Real Exchange Rates”. *The Economic Journal* 117 (2007), 1336–1356.
- Pascali, L. “The Wind of Change: Maritime Technology, Trade, and Economic Development”. *American Economic Review* 107 (Sept. 2017), 2821–2854.
- Reenen, J. Van. “Brexit’s Long-run Effects on the UK Economy”. *Brookings Papers on Economic Activity* (2016), 367–383. DOI: [10.1353/eca.2016.0031](https://doi.org/10.1353/eca.2016.0031).
- Sampson, T. “Brexit: The Economics of International Disintegration”. *Journal of Economic Perspectives* 31 (Sept. 2017), 163–184. DOI: [10.1257/jep.31.4.163](https://doi.org/10.1257/jep.31.4.163).

- Shiue, C. and W. Keller. “Markets in China and Europe on the Eve of the Industrial Revolution”. *American Economic Review* 97 (2007), 1189–1216. DOI: [10.1257/aer.97.4.1189](https://doi.org/10.1257/aer.97.4.1189).
- Steinberg, J. B. “Brexit and the Macroeconomic Impact of Trade Policy Uncertainty”. *Journal of International Economics* 117 (Mar. 2019), 175–195. DOI: [10.1016/j.jinteco.2019.01.009](https://doi.org/10.1016/j.jinteco.2019.01.009).
- Tintelnot, F. “Global Production with Export Platforms”. *The Quarterly Journal of Economics* 132 (2016), 157–209. DOI: [10.1093/qje/qjw037](https://doi.org/10.1093/qje/qjw037).
- Wong, W. F. “The Round Trip Effect: Endogenous Transport Costs and International Trade”. *American Economic Journal: Applied Economics* 14 (2022), 127–166. DOI: [10.1257/app.20190721](https://doi.org/10.1257/app.20190721).

Online Appendix

A Data and motivating evidence

A.1 Data construction

Table A.1: Store selection

Variable	Overall	BEL	FRA	UK
Transactions (count)				
·Included	7,662,402	1,040,485	3,524,775	3,097,142
·Excluded	276,764	49,278	126,511	100,975
Transactions (share)				
·Included	0.965	0.955	0.965	0.968
·Excluded	0.035	0.045	0.035	0.032
Liters (liters)				
·Included	52,354,103	9,541,668	32,302,062	10,510,372
·Excluded	1,631,999	543,285	877,058	211,655
Liters (share)				
·Included	0.970	0.946	0.974	0.980
·Excluded	0.030	0.054	0.026	0.020

Notes: This table shows the sample selection in terms of stores we include. We show for each country separately and overall how many transactions and consumed volume in liters we include and exclude.

Table A.2: Product selection

Variable	Overall	BEL	FRA	UK
Transactions (count)				
·Included	7,318,542	1,014,102	3,292,681	3,011,759
·Excluded	620,624	75,661	358,605	186,358
Transactions (share)				
·Included	0.922	0.931	0.902	0.942
·Excluded	0.078	0.069	0.098	0.058
Liters (liters)				
·Included	51,294,884	9,423,531	31,517,680	10,353,673
·Excluded	2,691,218	661,422	1,661,440	368,355
Liters (share)				
·Included	0.950	0.934	0.950	0.966
·Excluded	0.050	0.066	0.050	0.034

Notes: This table shows the sample selection in terms of which products we include. We show for each country separately and overall how many transactions and consumed volume in liters we include and exclude.

Table A.3: Hedonic regression

$\ln(p_{i,dt})$	Raw sample		Clean sample	
	(1)	(2)	(3)	(4)
$\mathbb{1}(\text{Sparkling})_j$	0.057 (0.040)	0.0159 (0.026)	0.084* (0.044)	0.0216 (0.030)
$\mathbb{1}(\text{Flavored})_j$	0.527*** (0.046)	0.569*** (0.032)	0.484*** (0.055)	0.555*** (0.044)
$\mathbb{1}(\text{Glass bottle})_j$	0.434*** (0.112)	0.336*** (0.072)	0.252 (0.237)	0.31** (0.141)
$\mathbb{1}(\text{Other package})_j$	0.456*** (0.060)	0.104 (0.101)	0.484*** (0.068)	0.0331 (0.121)
$\mathbb{1}((750\text{ml}, 1500\text{ml}))_j$	-0.401*** (0.051)	-0.456*** (0.029)	-0.359*** (0.058)	-0.434*** (0.035)
$\mathbb{1}(\geq 1500\text{ml})_j$	-0.965*** (0.053)	-0.893*** (0.027)	-0.918*** (0.056)	-0.877*** (0.034)
$\mathbb{1}(\text{Private label})_j$	-0.4*** (0.040)		-0.426*** (0.046)	
Region-Time FEs	✓	✓	✓	✓
Brand FEs		✓		✓
$Adj.R^2$	0.60	0.79	0.65	0.82
No. obs	546,972	546,972	370,031	370,031

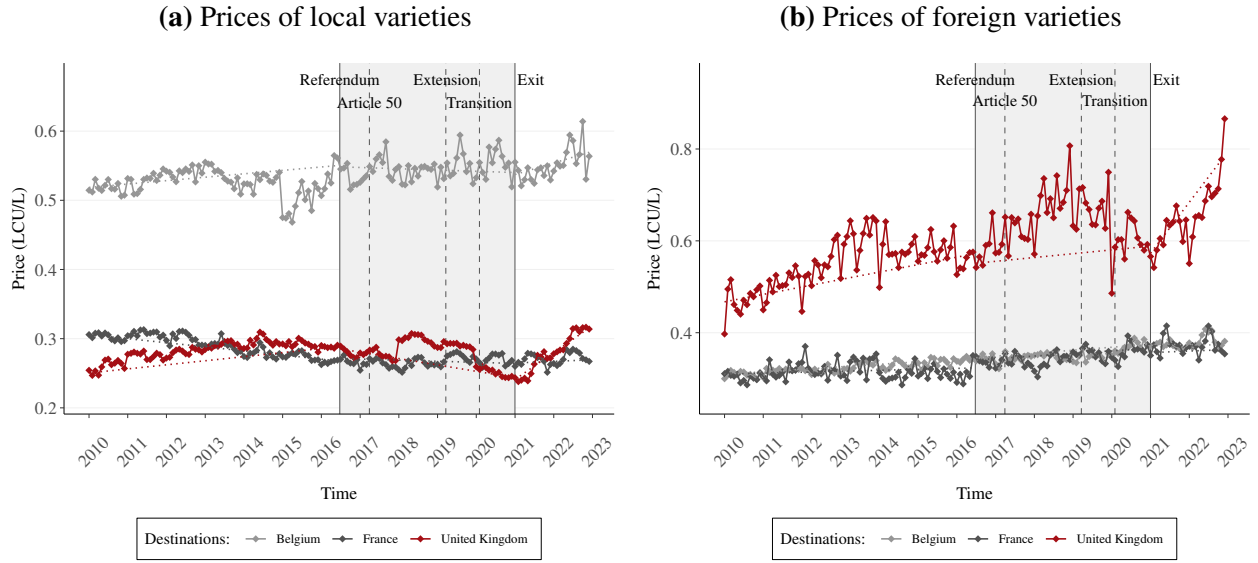
Notes: This table provides the results from regressing tax-inclusive consumer prices at the variety-NUTS-quarterly level on product characteristics. We aggregate transaction-level prices from the transaction level to the variety level using population weights interacted with volume sold to ensure that prices are representative. We cluster standard errors at the variety level and report significance levels at the $p < 0.1^*$, $p < 0.05^{**}$ and $p < 0.01^{***}$ levels.

Table A.4: Product characteristics: R^2

$p_{j,lt}$	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
PANEL (A): RAW DATA									
R^2	0.455	0.455	0.531	0.505	0.664	0.624	0.700	0.708	0.708
N	959,024	959,024	959,024	959,024	959,024	959,024	959,024	959,024	959,024
No. FEs	4,348	4,350	4,587	4,352	4,406	4,355	4,424	4,427	4,427
PANEL (B): CLEANED SAMPLE									
R^2	0.446	0.446	0.534	0.503	0.694	0.653	0.735	0.743	0.743
N	730,476	730,476	730,476	730,476	730,476	730,476	730,476	730,476	730,476
No. FEs	3,960	3,962	4,103	3,964	3,987	3,967	4,004	4,009	4,008
·Region $\times$ Time	✓	✓	✓	✓	✓	✓	✓	✓	✓
·Brand	✓	✓	✓	✓	✓	✓	✓	✓	✓
·Carbonated		✓	✓	✓	✓	✓	✓	✓	✓
·Flavor			✓						
·1 (Flavored)				✓	✓	✓	✓	✓	✓
·Bottle size					✓		✓	✓	✓
·Binned Bottle size						✓			
·Bottles per pack							✓	✓	✓
·Package type								✓	
·Binned Package type									✓

Notes: This table provides the results from regressing tax-inclusive consumer prices on fixed effects capturing various levels of variation in the data. To conduct these estimations, I show the resulting adjusted R^2 , the number of observations included in the regression, which of the fixed effects are included and the total number of included fixed effects.

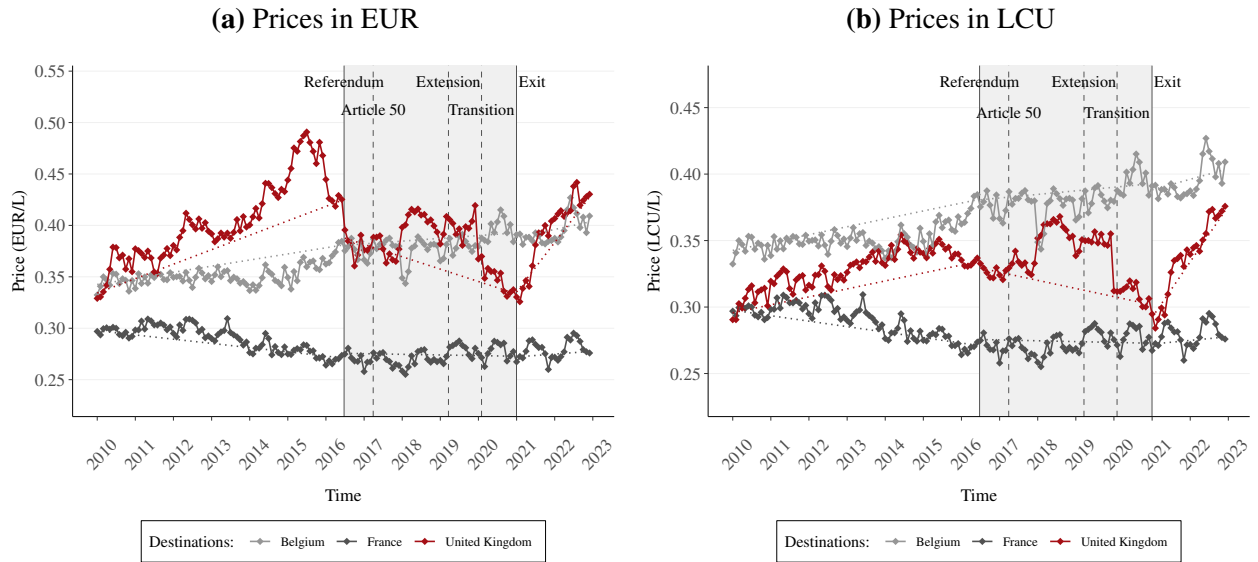
Figure A.1: Evolution of consumer prices



Notes: This figure plots the aggregate evolution of consumer prices for bottled water across Belgium, France, and the UK expressed in local currency units at a monthly frequency separately for foreign and local varieties. We define local varieties as varieties that are sourced in the same country as they are consumed; foreign varieties are defined as the antecedent to local varieties. We compute these series by aggregating over households, stores and bottled water varieties by weighting transaction-level unit prices using the interaction of population weights and volumes.

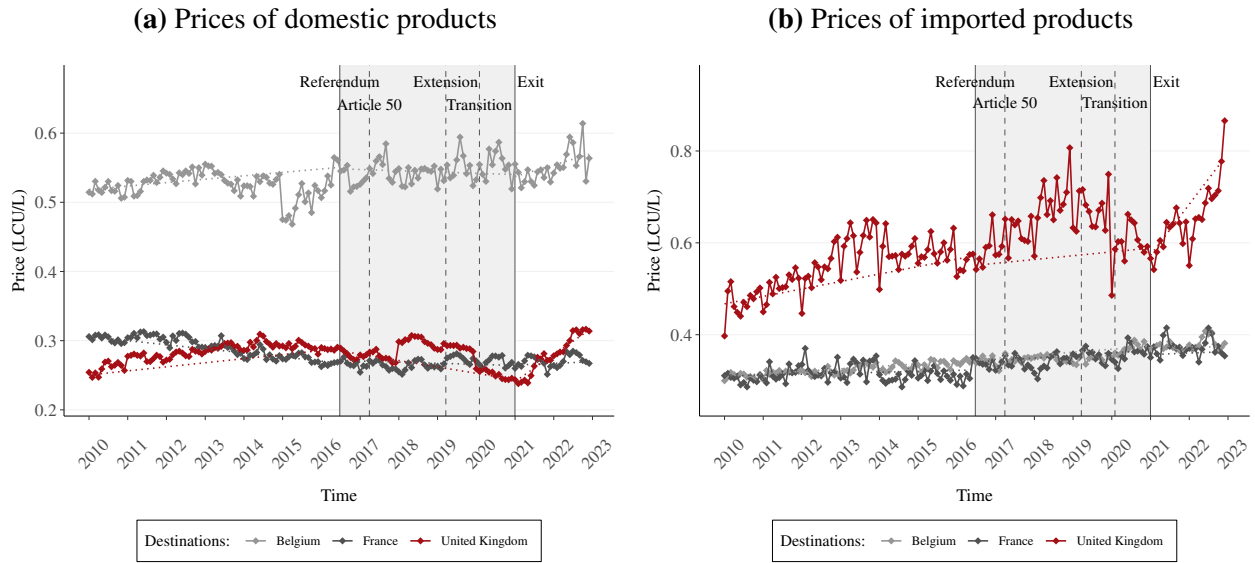
A.2 Evolution of consumer prices

Figure A.2: Evolution of consumer prices by currency



Notes: This figure plots the aggregate evolution of consumer prices for bottled water across Belgium, France, and the UK. Panel (A) expresses prices in EUR and Panel (B) expresses prices in local currency units. We compute these series by aggregating over households, stores and bottled water varieties by weighting transaction-level unit prices using the interaction of population weights and volumes.

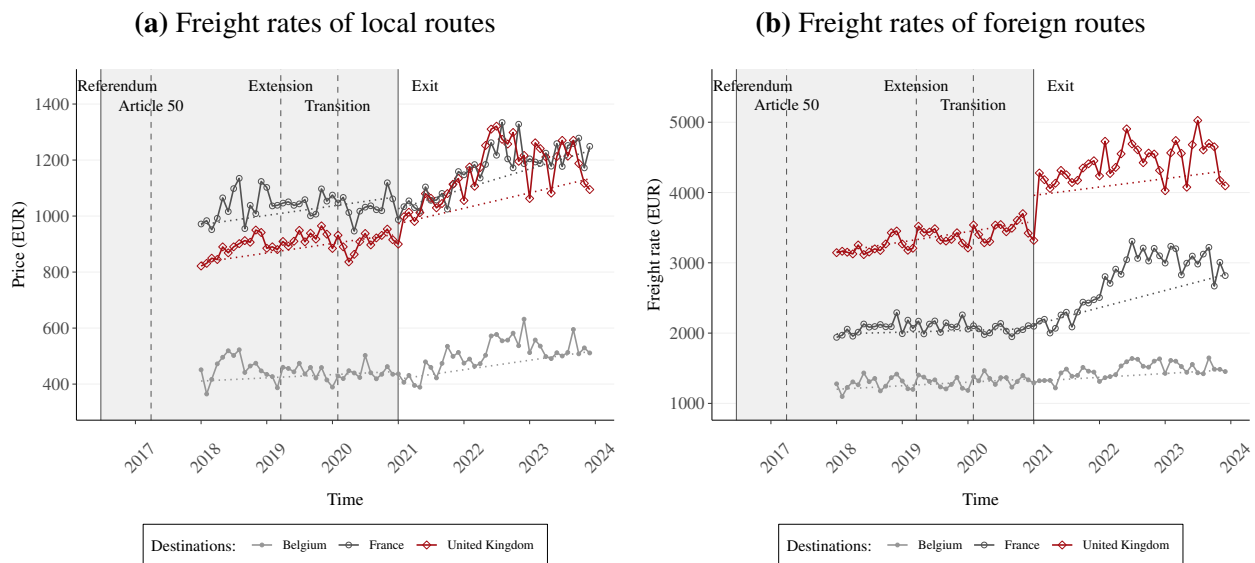
Figure A.3: Evolution of consumer prices by origin



Notes:

A.3 Evolution of freight rates

Figure A.4: Evolution of freight rates

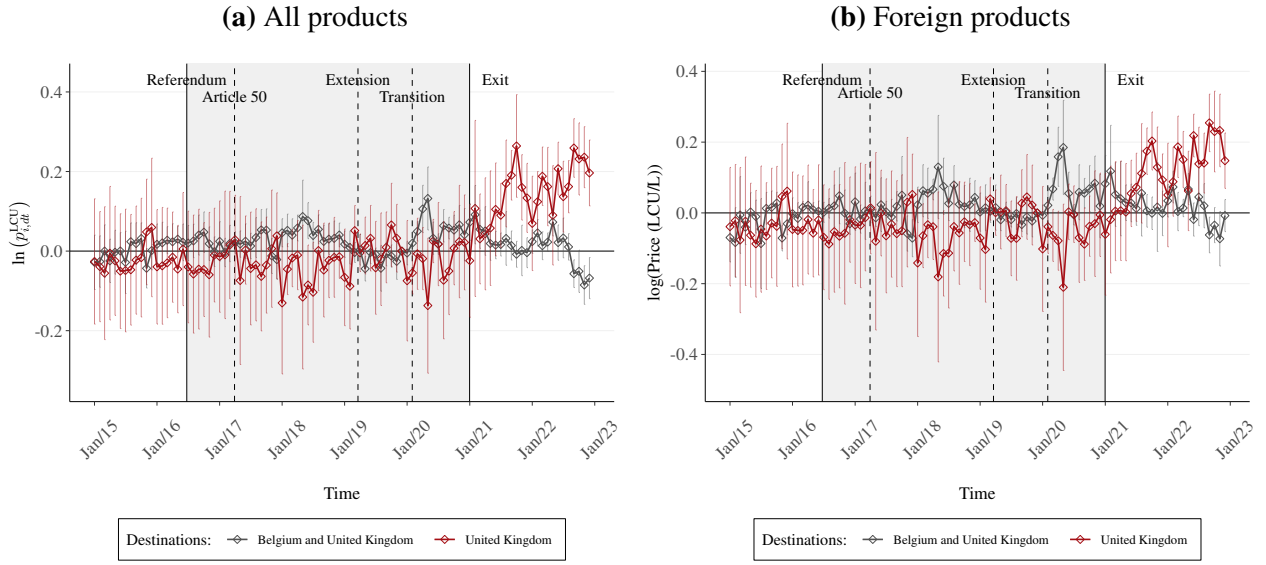


Notes:

B Additional results Reduced-form results

B.1 Consumer prices

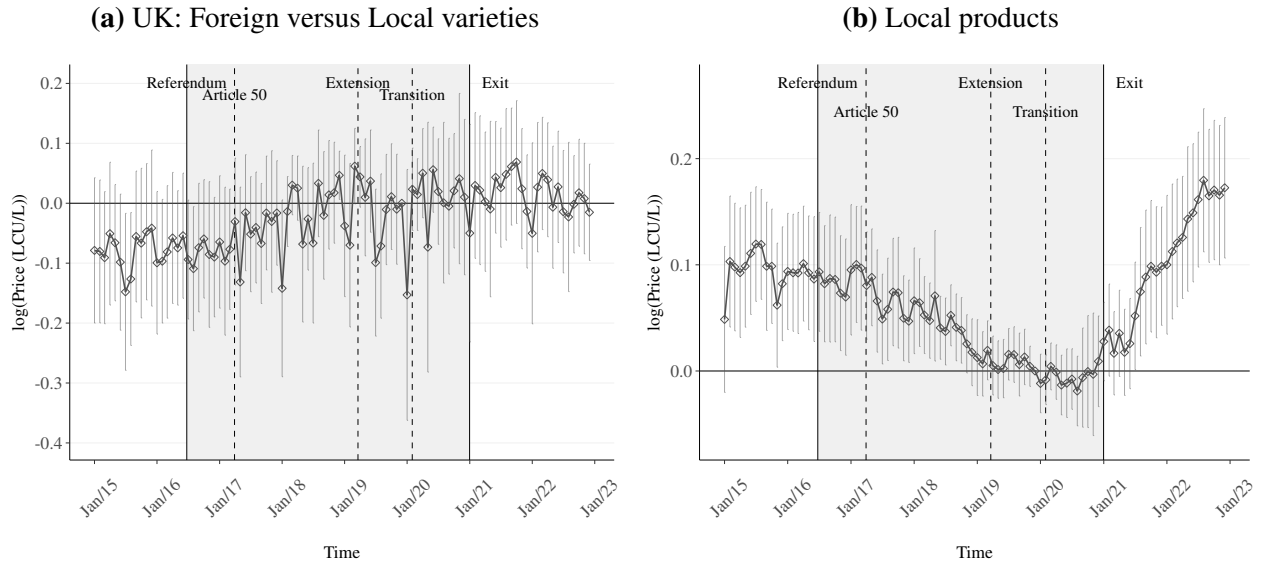
Figure B.1: Triple Difference-in-difference



Notes: We estimate the following model:

$$\ln(p_{i,dt}) = \sum_t \beta_t^1 (T_{i,d}^1 \times \mathbf{1}(t \neq 2019M12)) + \sum_t \beta_t^2 (T_{i,d}^2 \times \mathbf{1}(t \neq 2019M12)) + \theta_{i,d} + \lambda_{i,t} + \varepsilon_{i,dt}$$

Figure B.2: Difference-in-difference - Consumer prices



Notes: We estimate the following model:

$$\ln(p_{i,dt}) = \sum_t \beta_t (T_{i,d} \times \mathbf{1}(t \neq 2019M12)) + \theta_{i,d} + \lambda_t + \varepsilon_{i,dt}$$

Table B.1: Difference-in-differences - Consumer prices - Population weights

	All			Foreign	Placebo	Domestic DiD	Local
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$\mathbb{1}[d = UK] \times \mathbb{1}[t \leq 2015]$	0.011 (0.0236)	0.0109 (0.0229)	-0.00771 (0.0643)	-0.0264 (0.067)	-0.0751*** (0.0208)	-0.0244 (0.0403)	0.0331 (0.0302)
$\mathbb{1}[d = UK] \times \mathbb{1}[t = 2016]$	0.0301* (0.0165)	0.0301* (0.0162)	-0.0182 (0.0401)	-0.0271 (0.0424)	0.0256* (0.0136)	-0.031 (0.0294)	0.0553** (0.0242)
$\mathbb{1}[d = UK] \times \mathbb{1}[t = 2017]$	0.0275** (0.0132)	0.0283** (0.0131)	-0.0261 (0.0335)	-0.0295 (0.0363)	0.0195 (0.0148)	-0.0304 (0.0199)	0.057*** (0.0171)
$\mathbb{1}[d = UK] \times \mathbb{1}[t = 2018]$	0.0193** (0.0081)	0.0204** (0.00803)	-0.0353** (0.0157)	-0.0342* (0.0179)	0.0434** (0.0183)	-0.00951 (0.0105)	0.0325*** (0.00993)
$\mathbb{1}[d = UK] \times \mathbb{1}[t = 2019]$	-	-	-	-	-	-	-
$\mathbb{1}[d = UK] \times \mathbb{1}[t = 2020]$	-0.00377 (0.018)	-0.000669 (0.0179)	0.000666 (0.0122)	0.00404 (0.0149)	0.0603*** (0.0138)	-0.00851 (0.0181)	-0.0133 (0.0116)
$\mathbb{1}[d = UK] \times \mathbb{1}[t = 2021]$	0.0703*** (0.0265)	0.0652** (0.0267)	0.113*** (0.0209)	0.104*** (0.0227)	0.0354* (0.0193)	0.0352 (0.0334)	0.038 (0.0307)
$\mathbb{1}[d = UK] \times \mathbb{1}[t = 2022]$	0.187*** (0.0278)	0.181*** (0.0281)	0.166*** (0.036)	0.169*** (0.039)	0.0534** (0.0213)	-0.0108 (0.0375)	0.154*** (0.0327)
$\mathbb{1}[d = UK]$	-0.231*** (0.0685)	-	-	-	-	-	-
θ_i	✓						
$\theta_{i,d}$		✓	✓	✓	✓	✓	✓
λ_t	✓	✓				✓	✓
$\lambda_{i,t}$			✓	✓	✓		
R^2	0.029	0.023	0.009	0.067	0.345	0.004	0.024
No. obs	7,355	7,355	7,355	1,104	786	1,605	3,776

Notes: This table shows the results from estimating Equation (1) for different sets of controls and definitions of the treatment group using data at the yearly frequency. Columns (1)-(3) include both locally produced UK and varieties imported into the UK. In addition, column (1) includes variety and time fixed effects, column (2) includes variety-destination and time fixed effects and column (3) includes variety-destination and variety-time fixed effects. Column (4) includes variety-destination and variety-time fixed effects but focuses exclusively on varieties imported into the UK. Column (5) considers the same sample as column (5) but excludes UK consumer prices and considers a placebo experiment by defining the treatment group as consumer prices in Belgium. Columns (6) and (7) both include variety-destination and time fixed effects but differ in the sample and the definition of the treatment group. Column (6) only includes UK consumer prices on locally produced UK and varieties imported into the UK and defines the treatment group as UK consumer prices of imported varieties. Column (7) only includes consumer prices of locally sourced varieties in Belgium, France and the UK and defines the treatment group as consumer prices of locally sourced UK varieties. We estimate each specification using a weighted OLS estimator where the weights are population weights and the volume sold. We cluster standard errors at the variety level and denote statistical significance at the $p < 0.1^*$, $p < 0.05^{**}$ or $p < 0.01^{***}$ level.

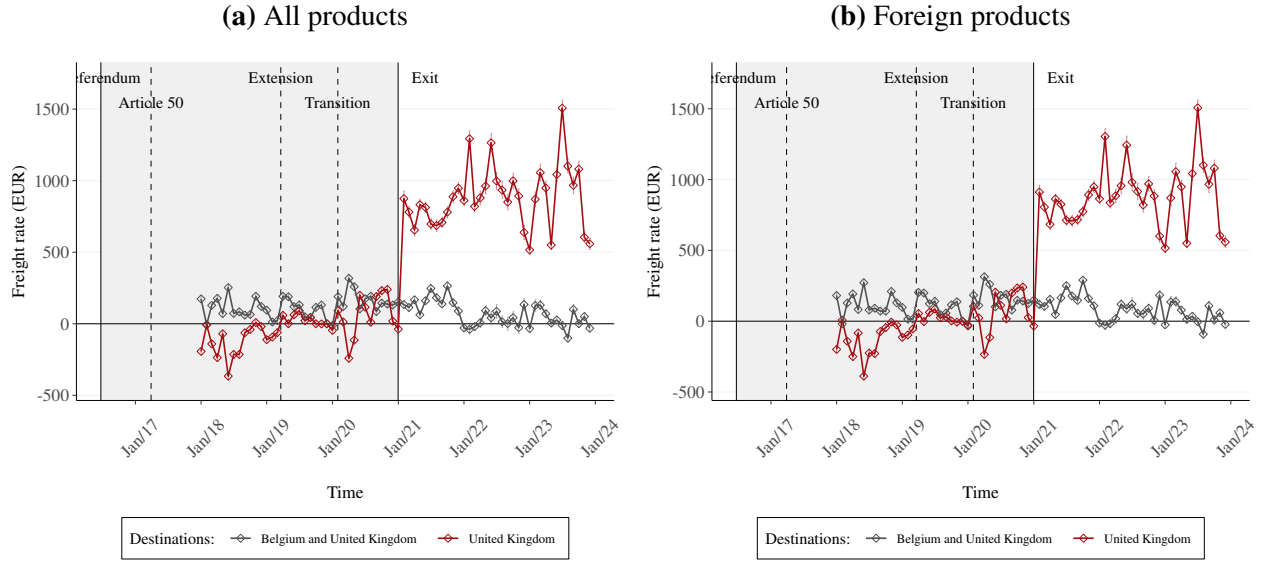
Table B.2: Difference-in-differences - Consumer prices - No weights

	All			Foreign	Placebo	Domestic DiD	Local
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$T_{i,d} \times \mathbb{1}(t \leq 2015)$	0.0812*** (0.0301)	0.0636** (0.0292)	0.0957 (0.069)	0.0873 (0.063)	-0.0992** (0.0381)	-0.00146 (0.0595)	0.0534 (0.0373)
$T_{i,d} \times \mathbb{1}(t = 2016)$	0.0137 (0.0231)	0.0142 (0.0223)	0.000933 (0.0494)	0.0259 (0.0561)	0.00603 (0.038)	-0.0523 (0.0462)	0.0374 (0.0285)
$T_{i,d} \times \mathbb{1}(t = 2017)$	0.022 (0.026)	0.0187 (0.0255)	0.0435 (0.0415)	0.0713 (0.0458)	-0.0311 (0.0501)	-0.0256 (0.0457)	0.0263 (0.034)
$T_{i,d} \times \mathbb{1}(t = 2018)$	0.00154 (0.0194)	0.00157 (0.0191)	0.00401 (0.035)	0.0376 (0.0332)	-0.00831 (0.0359)	-0.000854 (0.037)	0.00469 (0.025)
$T_{i,d} \times \mathbb{1}(t = 2019)$	-	-	-	-	-	-	-
$T_{i,d} \times \mathbb{1}(t = 2020)$	0.0273 (0.0251)	0.0203 (0.0249)	-0.00832 (0.0434)	0.0201 (0.0348)	0.0749 (0.0488)	-0.00881 (0.0406)	0.027 (0.0352)
$T_{i,d} \times \mathbb{1}(t = 2021)$	0.0144 (0.0394)	0.0141 (0.0396)	0.06 (0.0513)	0.0812 (0.0548)	0.0296 (0.0373)	0.0618 (0.0603)	-0.0174 (0.0548)
$T_{i,d} \times \mathbb{1}(t = 2022)$	0.0893** (0.0453)	0.0816* (0.0455)	0.204*** (0.0491)	0.197*** (0.0446)	0.0454 (0.0391)	0.102 (0.0694)	0.0432 (0.0632)
$T_{i,d}$	-0.156** (0.061)	-	-	-	-	-	-
θ_i	✓						
$\theta_{i,d}$		✓	✓	✓	✓	✓	✓
λ_t	✓	✓				✓	✓
$\lambda_{i,t}$			✓	✓	✓		
R^2	0.011	0.004	0.010	0.026	0.133	0.004	0.004
No. obs	7,355	7,355	7,355	1,104	786	1,605	3,776

Notes: This table shows the results from estimating Equation (1) for different sets of controls and definitions of the treatment group using data at the yearly frequency. Columns (1)-(3) include both locally produced UK and varieties imported into the UK. In addition, column (1) includes variety and time fixed effects, column (2) includes variety-destination and time fixed effects and column (3) includes variety-destination and variety-time fixed effects. Column (4) includes variety-destination and variety-time fixed effects but focuses exclusively on varieties imported into the UK. Column (5) considers the same sample as column (5) but excludes UK consumer prices and considers a placebo experiment by defining the treatment group as consumer prices in Belgium. Columns (6) and (7) both include variety-destination and time fixed effects but differ in the sample and the definition of the treatment group. Column (6) only includes UK consumer prices on locally produced UK and varieties imported into the UK and defines the treatment group as UK consumer prices of imported varieties. Column (7) only includes consumer prices of locally sourced varieties in Belgium, France and the UK and defines the treatment group as consumer prices of locally sourced UK varieties. We estimate each specification using an unweighted OLS estimator. We cluster standard errors at the variety level and denote statistical significance at the $p < 0.1^*$, $p < 0.05^{**}$ or $p < 0.01^{***}$ level.

B.2 Freight rates

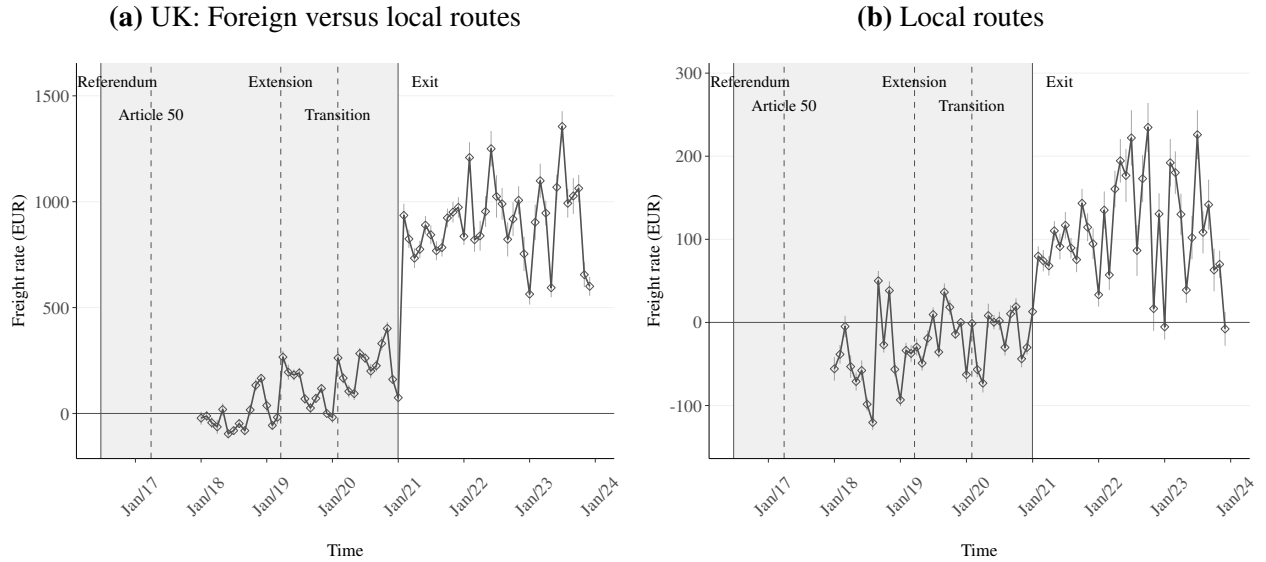
Figure B.3: Triple Difference-in-difference - Freight rates



Notes: We estimate the following model:

$$\ln(p_{od,t}) = \sum_t \beta_t^1 (T_{od}^1 \times \mathbb{1}(t \neq 2019M12)) + \sum_t \beta_t^2 (T_{od}^2 \times \mathbb{1}(t \neq 2019M12)) + \theta_{od} + \lambda_{o,t} + \varepsilon_{od,t}$$

Figure B.4: Difference-in-difference - Freight rates



Notes: We estimate the following model:

$$\ln(p_{od,t}) = \sum_t \beta_t (T_{od} \times \mathbb{1}(t \neq 2019M12)) + \theta_{od} + \lambda_t + \varepsilon_{od,t}$$